

AI – DRIVEN TRAFFIC LIGHT AUTOMATION

A Dissertation submitted in partial fulfillment of the requirements for the award of
degree of

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS & COMMUNICATIONS ENGINEERING

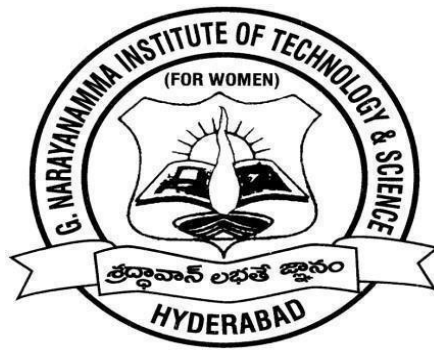
Submitted By

N. BHAVYA SREE	22251A0423
K. SAI SHIVANI	22251A0440
P. NITHYANJALI	22251A0453
A. SRIHITHA	22251A0403

Under the esteemed guidance of

Mrs. M. Shanthi

Asst. Professor, ECE, GNITS.



DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGG
G. NARAYANAMMA INSTITUTE OF TECHNOLOGY & SCIENCE

(For Women)

(AUTONOMOUS)

(Accredited by NBA & NAAC)

2025-26

DEPT. OF ELECTRONICS & COMMUNICATION ENGG

CERTIFICATE

THIS IS TO CERTIFY THAT THE MAJOR PROJECT ENTITLED

AI – DRIVEN TRAFFIC LIGHT AUTOMATION

IS THE BONAFIDE WORK OF

N. BHAVYA SREE	22251A0423
K. SAI SHIVANI	22251A0440
P. NITHYANJALI	22251A0453
A. SRIHITHA	22251A0403

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENT FOR THE
AWARD OF DEGREE OF BACHELOR OF TECHNOLOGY IN ELECTRONICS AND
COMMUNICATION ENGINEERING DURING THE YEAR 2025-26.

GUIDE

Mrs. M. Shanthi
Asst. Professor, ECE

Dr. K. Ragini

Prof & HOD - ECE

EXTERNAL EXAMINER

ACKNOWLEDGEMENTS

We are extremely thankful and indebted to our project guide and project coordinator, **Mrs. M. Shanthi**, Assistant Professor, Department of ECE, GNITS for her constant guidance, encouragement, and moral support throughout the project.

Our sincere thanks and gratitude to **Dr. K. Ragini**, Professor & Head, Department of ECE, GNITS, for all the timely support and valuable suggestions during the period of our project.

We would like to express our sincere thanks to **Dr. K. Ramesh Reddy**, Principal, GNITS, for providing the working facilities in the college.

Finally, we are extremely thankful to all the faculty members and staff of the ECE Department who helped us directly or indirectly, parents and friends for their cooperation in completing the project work.

N. BHAVYA SREE	22251A0423
K. SAI SHIVANI	22251A0440
P. NITHYANJALI	22251A0453
A. SRIHITHA	22251A0403

INDEX

S. No	Topic	Page. No
	Abstract	i
	List of Figures	ii
	Acronyms	iii
1	Introduction	
	1.1 Background	1
	1.2 Disadvantages of Conventional Traffic Signal System	2
	1.3 Brief Overview of the topic	3
	1.4 Importance and Relevance	4
	1.5 Methodology	5
	1.6 Vision for the Future	6
	1.7 Societal Impact	6
	1.8 Conclusion	7
2	Literature Survey	
	2.1 Introduction	8
	2.2 Traditional Traffic control System	8
	2.3 Traffic Detection based on Deep Learning with YOLO	10
3	Research Rational and Objectives	
	3.1 Problem Statement	11
	3.2 Proposed System Motivation	12
	3.3 Objectives	12
	3.4 Project Scope and Limitations	13
	3.4.1 Scope of the work	13
	3.4.2 Limitations	14
4	Proposed System	

	4.1 Introduction	17
	4.2 Introduction to the Proposed System	18
	4.3 Elements of the Proposed System	19
	4.4 Workflow of the Proposed System	20
	4.5 Benefits of the Proposed System	21
	4.6 System Design Methodology	21
	4.7 Data flow of the Proposed System	22
	4.8 Priority of Emergency Vehicles	23
5	System Architecture	
	5.1 Introduction	25
	5.2 Block Diagram	27
	5.3 Functional Models of the System	30
	5.4 Hardware and Software Integration	33
	5.5 Data Processing	36
6	Hardware Components	
	6.1 Introduction	39
	6.2 Raspberry Pi Zero 2W	40
	6.3 Pi Camera Module	41
	6.4 PIC Microcontroller	43
	6.5 ULN2803 Driver IC	45
	6.6 Stepper Motor	47
	6.7 Traffic Signal LEDs	49
	6.8 Crystal Oscillator	50
	6.9 Regulated Power Supply	52
7	Software Implementation	
	7.1 Introduction	54
	7.2 Programming Language- Python	55
	7.3 YOLO Object Detection Algorithm	56
	7.4 OpenCV Image Processing	56

	7.5 Vehicle Detection Process	57
	7.6 Calculation of the density of traffic	57
	7.7 Serial Communication with Microcontroller	58
8	Circuit Diagram and Hardware Implementation	
	8.1 Introduction	59
	8.2 The description of Circuit Diagram	60
	8.3 Hardware Implementation	62
	8.4 System Integration	63
9	Working Principle	
	9.1 Introduction	64
	9.2 Traffic Image Capture	65
	9.3 Automobile Detection with YOLO	66
	9.4 Traffic Density Estimation	69
	9.5 Traffic Signal Control	70
	9.6 Camera Rotation Mechanism	72
10	Results and Analysis	
	10.1 Introduction	74
	10.2 Vehicle Detection results	74
	10.3 Traffic Density Analysis	75
	10.4 Traffic Signal Operation	76
	10.5 System Performance Evaluation	78
11	Conclusion	
	References	81

ABSTRACT

Traffic jam has turned out to be one of the most significant problems of contemporary cities following the fast population increase and the growing quantity of vehicles in the street. Traditional traffic regulation systems are largely based on fixed-time traffic lights, which work on the duration of already-programmed time interval regardless of actual traffic conditions. Consequently, these systems can tend to create inefficient traffic flow, longer waiting time, increased fuel consumption and high air pollution. Moreover, conventional systems do not give special consideration to the emergency vehicles like ambulances, fire engines and police vehicles which can cause life threatening delays in the emergency cases.

The system consists of a Raspberry Pi Zero 2W which serves as a central processing unit and a Pi Camera to give live video of the traffic in the road intersections. The frames of the video taken are analyzed through the YOLO (You Only Look Once) object detection algorithm, which is known to be very accurate and is capable of running with real time and detecting several objects in a frame.

The YOLO model recognizes and labels various classes of vehicles such as cars, motorcycles and buses, trucks, and emergency vehicles. The system uses the information about the bounding box that the detection model produces and counts the quantity of vehicles in every lane in order to estimate the density of the traffic. The system classifies the traffic conditions based on fixed density levels of low, medium and high traffic density. The system changes the traffic signal length dynamically per the identified traffic density to help in facilitating a better flow of traffic and minimize any congestion at crossways.

LIST OF FIGURES

S. No	Fig No	Name of the fig	Page No
1	5.1	Block Diagram	27
2	6.1	Raspberry Pi Zero 2w	40
3	6.2	Pi Camera	42
4	6.3	PIC Microcontroller	44
5	6.4	ULN2803 Driver IC	46
6	6.5	Stepper Motor	47
7	6.6	Traffic Signal LEDs	49
8	6.7	Crystal oscillator	51
9	8.1	Circuit Diagram	60
10	10.1	Detection of Emergency Vehicle	75
11	10.2	Detection of Cars	75
12	10.3	Low traffic density with green signal duration of 6s	76
13	10.4	High traffic density with green signal duration of 12s	76
14	10.5	Hardware setup of traffic system	77
15	10.6	Working model of traffic system	77
16	10.7	Confusion Matrix	78
17	10.8	F1-Confidence Curve	78
18	10.9	Precision-Recall Curve	78
19	10.10	Precision-Confidence Curve	78

ACRONYMS

S. No	Abbreviations	Full form
1	AI	Artificial Intelligence
2	YOLO	You Only Look Once
3	PIC	Programmable Interface Controller
4	GPIO	General Purpose Input Output
5	LED	Light Emitting Diode
6	IoT	Internet of Things
7	IR	Infrared
8	OpenCV	Open Source Computer vision Library
10	MQTT	Message Queuing Telemetry Transport
11	CPU	Central Processing Unit
12	IC	Integrated Circuit

CHAPTER 1

INTRODUCTION

1.1 Background

Transportation is a critical aspect of the modern society because it facilitates the transportation of people and goods between one location and the other. As the population grows and cities populate, the traffic in the road has grown substantially in the last several decades. This high rate of vehicular traffic growth has led to severe problem of traffic congestion, more particularly in the metropolitan and urban regions. Traffic jams do not only result in delays among the commuters but also release more fuel and pollute the environment as well as cost the economy.

The classic traffic control systems are largely premised on fixed time intervals in which the traffic lights vary based on the given timing patterns. These systems are not mindful of the traffic conditions that are available at the intersection. As an illustration, the same duration of the green signal can be given to a lane that has few vehicles as compared to a congested lane. This is an ineffective distribution of signals, that leads to unnecessary waiting time as well as ineffective traffic management.

In addition, there is also a high likelihood of delays in traffic intersections of the emergency vehicles like the ambulances, fire engines and the police vehicles since there are no intelligent traffic management systems. These delays may be disastrous particularly when there is a critical medical emergency or a disaster. Thus, there exists an immediate necessity to come up with the intelligent traffic management systems, which will be able to keep track of traffic conditions and adjust signal times on the fly.

The development of the sphere of artificial intelligence, computer vision, and embedded systems has established new possibilities of the intelligent management of traffic. Traffic density can be analyzed and traffic lights can be controlled more effectively by using cameras and machine-learning algorithms. Urban transportation systems can be increased by a great number with the help of these technologies and help to create smart cities.

1.2 Disadvantages of the Conventional Traffic Signal System

The traditional traffic light systems are run under fixed time mechanism which is preprogrammed and never adjusts to the real-time traffic situations. One of the main limitations of such systems is that despite the simplicity and ease of implementation, they have a few significant limitations that influence the efficiency of the traffic overall.

A major weakness of the traditional traffic lights is their ineffectiveness in responding to the variation in traffic density. The heavy traffic congestion may occur on some lanes during peak hours with some lanes being relatively empty. Fixed time signal systems however provide equal signal durations to each of the directions irrespective of the real load of traffic. Consequently, cars in traffic jam lanes spend much time waiting and this causes frustration among road users and low level of efficiency in the road.

The other major disadvantage is that there is no preference given to the emergency vehicles. In the traditional systems, the traffic lights switch based on time schedules which do not identify the availability of ambulances, fire brigade or police cars. It can postpone the services of emergency response that may lead to life-threatening cases.

Of, what is more, the traditional traffic systems are not able to track the real-time conditions on the roads, including accidents, congestion, stalled vehicles, or abnormal congestion patterns. The traffic authorities are not able to respond promptly to unexpected happenings on the road without constant monitoring.

Moreover, most of the conventional traffic monitoring tools are based on hardware devices including inductive loops or infrared sensors in the road surface. These types of sensors are not cheap to put up and take care of, and they are subject to failure because of the environment or damage.

As a result of the latter, the current transportation systems need more sophisticated and smarter mechanisms of traffic control capable of processing the real-time traffic data and making dynamic decisions to enhance the traffic flow and road safety.

1.3 Brief Overview of the Topic

The increasing problems relating to traffic congestion have led to the emphasis in the intelligent traffic management systems capable of monitoring and controlling traffic in a more effective and responsive way. Intelligent traffic systems involve the application of new technologies to enhance the monitoring of the traffic and controlling of signals by using artificial intelligence, machine learning, image processing, and Internet of Things devices.

The first important benefit of intelligent traffic management systems is the possibility to study the current state of traffic in real time with the help of cameras and sophisticated algorithms. These systems are able to dynamically adjust signal timings depending on the vehicle density by consistently tracking the traffic flow and, thus, reduce congestion and enhance the efficiency of the traffic in general.

Emergency vehicles can also be detected and prioritized using intelligent systems. Traffic lights can use image recognition or communication technologies to identify emergency vehicles and thus, automatically give them a green runway to enable them to travel fast across the intersection.

The other significant advantage of smart traffic is that it reduces fuel and air pollution. The short not of time that vehicles spent at the traffic lights would mean that the use of fuel would be less and there will be fewer emissions, thus making the environment clean and healthy.

Moreover, the smart cities can be developed with the help of intelligent traffic management systems that enhance the introduction of advanced technologies in the urban infrastructure. These systems may be linked to centralized traffic control stations at which the traffic information can be utilized to plan the transportation networks in the long-term perspective and optimizing the networks. Thus, intelligent traffic control systems should be implemented to address the increasing traffic problems in the contemporary cities.

The suggested project will cover the creation of an AI-based automation system of traffic lights that employs computer vision and embedded systems to manage traffic lights

in real-time. The system combines Raspberry Pi Zero 2W, Pi Camera, PIC microcontroller, and traffic signal LEDs to imitate a smart intersection of the traffic.

The Pi Camera is the device of this system which records the real-time traffic at the junction. The Raspberry Pi processes these images with the help of the YOLO object detection algorithm to detect vehicles (cars, motorcycles, buses, trucks, and emergency vehicles). Judging by the identified objects, the system is able to compute the number of cars in each of the lanes and arrive at the density of the traffic.

The main processing unit will be the Raspberry Pi which will handle the processing of images and decision-making. The traffic density is then computed and control signals are relayed to the PIC microcontroller either by a serial or a GPIO connection. The PIC microcontroller will then turn the traffic lights LED to give the lanes that have higher traffic density a green light. Besides detection of traffic density, emergency vehicle is also detected in the system. An emergency vehicle is detected and the system gives the vehicle a green light to pass through the intersection fast.

The camera is rotated by the stepper motor thus allowing the system to view traffic in various directions. The motor stepper is connected to the microcontroller via the ULN2803 driver IC. Controlled power supply maintains a constant voltage on all the hardware components. Combining artificial intelligence and embedded hardware allows building a smart traffic control system that can adapt to the current real-time traffic conditions. The system manages traffic and ensures that there is minimal waiting time and also increases emergency response efforts.

1.4 Importance and Relevance

The traffic control is critically required in order to alleviate urban traffic to deliver better mobility, less pollution to the environment and to maximize human safety. The effects of traffic congestion are very detrimental to the society since it causes economic loss, financial usage of fuels and degradation of the environment.

Wastage of fuel and time are considered to be one of the greatest issues brought by congestion. Traffic jam vehicles waste money during idling in the traffic thereby adding to the fuel expenses of motorists and adding to the air pollution. The release of dangerous

gases including carbon dioxide (CO₂), carbon monoxide (CO), and nitrogen oxides (NO_x) in the atmosphere has a serious impact on the air quality and leads to global warming. Besides environmental issues, traffic jam is also an implication to the emergency response system. Ambulances and fire engines take long to make their way through busy intersections. Such delays will be disastrous to life threatening situations.

Introduction of a traffic light automation system based on AI will potentially have a greater positive impact on traffic control process as it will provide an opportunity to monitor active traffic in real time and adapt the signal to the traffic. The system will be able to optimize the traffic and cut on waiting time in intersections through analysis of live traffic data.

Moreover, the system can become significant in the formation of the smart cities. It is possible to implement intelligent traffic management systems with cloud platforms and Internet of Things (IoT) networks to allow centralized monitoring and data analysis. With the data collected, authorities can use the result to establish traffic Routes, Prejudge traffic jam, and devise the infrastructure refinements.

1.5 Methodology

The given traffic automation system has well-developed methodology based on the use of the technologies of computer vision and artificial intelligence, as well as embedded systems. The system starts by recording the live video of the traffic at the intersection with the help of camera. The frames of the captured video are analyzed through OpenCV, which carries out preprocessing activities of the image including frame extraction and image making possible. The processed frames are then sent to the YOLO object detection algorithm that provides information of the objects appearing in the scene and classifies them. The cars, trucks, buses, motorcycles, and emergency vehicles are identified and outlined with bounding boxes.

The system is used to measure the number of cars in a given lane in order to estimate the traffic density. Depending on the number of vehicles, the Raspberry Pi controller determines the right moment of time to give the signal to each lane.

The signal control logic identifies the lane to which the green signal is to be allocated

within a specific period of time. The software and hardware parts are brought to communicate with each other through the use of PySerial that transmits control commands to LEDs modeling traffic lights..

1.6 Vision for the Future

The system suggested can become something significant in the perspective of the smart cities of the future. With the ever increasing size of urban population, traffic management systems need to be smarter, dynamic and expansive. The system can also be combined with cloud-based platforms to allow centralized monitoring of traffic in the future through the intersection of many intersections. The information gained by various traffic cameras is capable of being compared to pinpoint the traffic trends and the congestion will be anticipated.

The system can also be connected to the autonomous vehicles, public transportation and smart navigation platforms. The historical traffic data can be used to train the machine learning models that will be able to predict the traffic conditions and thereby optimize the signal timing in advance. These smart traffic control systems will be influential in the efficient transportation in the cities and they will facilitate the sustainable development of urban centers.

1.7 Societal Impact

The system suggested can become something significant in the perspective of the smart cities of the future. With the ever increasing size of urban population, traffic management systems need to be smarter, dynamic and expansive.

The system can also be combined with cloud-based platforms to allow centralized monitoring of traffic in the future through the intersection of many intersections. The information gained by various traffic cameras is capable of being compared to pinpoint the traffic trends and the congestion will be anticipated.

The system can also be connected to the autonomous vehicles, public transportation and smart navigation platforms. The historical traffic data can be used to train the machine learning models that will be able to predict the traffic conditions and thereby optimize the

signal timing in advance. These smart traffic control systems will be influential in the efficient transportation in the cities and they will facilitate the sustainable development of urban centers.

1.8 Conclusion

Traffic management in contemporary urban setting is necessary to assure easy transportation and safety of people around. The conventional timer-based traffic signal controller is no longer adequate to deal with the dynamic and unpredictable nature of traffic in the cities.

The aim of this project is to provide an AI-based Traffic Light Automation System, where computer vision and machine learning methods would be used to observe and regulate traffic. The system can be used to dynamically set the traffic lights depending on the density of traffic and achieve the priority of the emergency vehicles by working with YOLO object detection algorithm and Raspberry Pi embedded hardware.

The offered system has a number of advantages as compared to the conventional traffic control systems, among which are, but are not limited to, better traffic efficiency, less time on the waiting scenario, decreased environmental impact, and better emergency response. Moreover, the system offers the basis of the future smart city traffic management solutions by uniting the artificial intelligence, real-time data processing, and embedded automation.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Traffic jamming has been a significant issue in cities because of high population growth rate and the number of vehicles in the roads. The conventional traffic control systems are run on a fixed schedule of timing, and they are not relevant to the real-time traffic situation. This has led to unnecessary delays at intersections by vehicles despite having minimal traffic at a particular lane. These inefficiencies cause the travel time to be more, fuel to be more consumed and pollution to the environment.

Conventional traffic control systems are largely founded on fixed-time based control patterns, in which the use of the signaling is founded on defined time durations without taking into account actual time traffic. Even though these systems are simple to operate, they usually result in unproductive traffic movements, particularly in the rush times when traffic volume is high. The fixed-timer systems are unable to dynamically change the signal timings depending on the situation in the traffic, and it leads to the unnecessary waiting time, more congestion, and waste of fuel.

In a bid to curb such constraints, scholars have tried different intelligent traffic management methods. Initial solutions were based on the relying of physical sensors in the road to notify the on-road presence of cars and manipulate the traffic lights. Although the sensor systems will be superior in offering improved traffic surveillance, like the traditional ones, it involves high installation and maintenance. Environmental factors such as rain, dust or road damage will also wreak havoc on sensors, this will lead to their breakdown and thus this will affect their reliability in long-term deployment [1].

2.2 Traditional Traffic Control Systems

The traditional traffic management systems mainly use fixed-time signal control systems. In such systems, traffic lights fulfill set time time frames irrespective of the quantity of vehicles awaiting at a crossroad. Although this method makes system design

less complicated, it lacks flexibility to real-time conditions in the traffic.

Maurya et al. [1] observe that in many cases fixed-time traffic signal systems result in inefficient traffic control particularly during peak hours when the density of vehicles across lanes is significantly varied. These systems can each system allot the same signal time to the lanes despite one lane in heavy traffic and another lane having very few vehicles. This will lead to increased waiting time and poor usage of the road infrastructure. Also, non-real-time systems will not be able to react to any unforeseen occurrences like accidents, vehicles malfunctions, and emergency vehicle movements. This has been translated into a continuous rise in traffic jams in most of the urban regions where the conventional signal systems are still operational.

In an effort to enhance the effectiveness of the traffic control measures, scholars presented sensor-driven systems, which identify the presence of vehicle through the deployment of gadgets like inductive loop sensors, infrared sensors, and pressure sensors that are placed within the surfaces of the roads. A similar traffic control system was proposed by Prakash and Thankappan [2], which involves image processing and sensor based detection to determine the density of the vehicles. Although these systems offer more precise traffic information than a timer-based system, they involve great infrastructure changes during implementation. Road sensors can also maintenance easily because of road wear and tear because of heavy traffic.

In another research study, Moka et al. [3] put forward a traffic surveillance system that integrated image processing and acoustic detection to detect emergency vehicles by using siren sounds. Even though the system tried to enhance the prioritization of emergency vehicles, it was challenged by problematic noisy urban areas where there are two sound sources and hence might not be detected correctly. Such restrictions illuminate that a more extensive and dependable traffic tracking approach is necessary, which is not that associated with physical sensors.

Computer vision and image processing have turned out to be a trending method of surveillance of traffic because it is flexible and scalable. The cameras mounted at intersections are able to record live a video feed of the conditions on the roads, the motion

of the vehicles on the road can be recorded and analyzed to get the density of the traffic, traffic flow and the level of congestion. Background subtraction method, frame differencing method, and edge detection method were some techniques which were used in processing images many years ago to detect moving cars. These methods however, typically had difficulty with changing lighting effects, shadows as well as not being able to see through cars due to overlapping.

2.3 Traffic Detection based on Deep Learning with YOLO

The recent progress in the field of deep learning has contributed to much higher object detection capabilities. In the deep learning algorithms, the YOLO (You Only Look Once) object detection model has received overwhelming features as it can run real-time object detection with high accuracy. A thorough review of the methods of identifying traffic signs and transportation tools through the use of the YOLO-based object detection technique has been discussed by Zabala-Blanco et al. [4]. The paper has also emphasized the benefits of YOLO models to use to perform image-processing operations on whole images in one sweep, which are much faster and efficient than traditional approaches.

Correspondingly, Sai Venu Prathap et al. [5] made an intelligent traffic light by an algorithm based on YOLO to identify vehicles and also determine the density of traffic. The study conducted by them showed that the accuracy of vehicle detection could be dramatically raised by using deep learning models, and dynamic control of traffic lights was possible.

Nguyen and Tran [7] suggested more recently a vehicle detection system based on traffic flow on the YOLOv8 algorithm in real-time. The system had a high detection rate and it was able to perform effective deep learning models in the analysis of traffic patterns in real time. Detection systems made using YOLO can detect various kinds of cars, buses, trucks, and motorcycles at the same time. The ability enables traffic management systems to better analyze the density of traffic as well as make decisions in signal management.

CHAPTER 3

RESEARCH RATIONALE AND OBJECTIVES

3.1 Problem Statement

Traffic congestion is now one of the most important issues in the city transportation systems. With more and more vehicles on the roads, the current traffic controlling mechanisms are failing to ensure that there is a smooth traffic flow. The traditional traffic signal systems are founded on the traditional fixed-time control systems which lack consideration of the real-time traffic conditions. This creates an ineffective operation of the traffic signals and unwarranted delays by cars at crossroads.

Lack of flexibility is one of the greatest issues related to the conventional traffic signal systems. These systems assign equal signal times to the lanes irrespective of the traffic density in the real sense on the lanes. As an illustration, a road with the few vehicles can be given the above green signal as compared to a road with heavy traffic jams. Such ineffective signal allocation will cause more waiting time and low efficiency in traffic.

The other significant problem is the lack of priority of emergency vehicles. Emergency vehicles like ambulances, fire trucks and police vehicles must have immediate access to intersections in traffic. The availability of such vehicles is however not detected by the traditional traffic lights which may lead to serious delays in case of an emergency. The situation with slow response time can be disastrous, especially in the case of a medical emergency where every second matters.

Besides the congestion and emergency vehicles delays, the conventional traffic system does not have the ability to identify an abnormal traffic situation like accidents, stalled traffic, or abnormal congestion patterns. Traffic authorities are not able to react to such situations in real-time when there is no real-time monitoring. Most of the available traffic monitoring systems use hardware sensors that are inbuilt in the road surfaces like inductive loop detectors or IR sensors. Even though they are capable of detecting the availability of vehicles, the installation and maintenance of these sensors is expensive.

3.2 Proposed System Motivation

The necessity to create the offered AI-based traffic light automation system is determined by the increasing demand to have more intelligent and smart traffic managerial strategies. Daily transportation is being affected by the growing level of traffic congestion that is being witnessed in cities as a result of the increasing number of people living in cities.

Traffic jam does not only lead to time wastage among commuters but also fuels and air pollution. Cars that take long durations to clinical traffic jams waste unnecessarily used fuel and pollute the atmosphere with poisonous gases. These adverse environmental effects can be minimized by enhancing the effectiveness of the traffic lights. The other key driving project is the inclination to facilitate emergency services. The ambulances and fire engines should be capable of operating at high speed across the traffic intersections to help in responding to an emergency. Smart traffic lights that give priority to emergency vehicles can help minimize the response time and enhance the safety of people greatly.

The development of new technologies in artificial intelligence and computer vision has allowed building traffic monitoring systems that involve cameras and deep learning algorithms to identify vehicles and analyze traffic patterns. YOLO object detection algorithm is also appropriate in real-time applications because it is fast in terms of detection framework and detection accuracy.

With the integration of artificial intelligence and embedded hardware platforms like Raspberry Pi and microcontrollers, one can create such a low-cost traffic management software that may be deployed in the real world. The suggested system is expected to use these technologies to develop a smart traffic control system, which would be able to adjust to the current actual traffic and enhance the overall traffic efficiency

3.3 Objectives

The primary aim of this project is to develop and establish an intelligent traffic signal control system, which will be able to change the timing of signal dynamically depending

on real-time traffic conditions. The system will enhance the efficiency of traffic by monitoring vehicle density and assigning signal periods respectively. Among the main goals of the project is to come up with a real-time vehicle detection system by the use of the YOLO object detection algorithm. The system will record the images of traffic with the help of a Pi Camera and analyze it with the help of the Raspberry Pi to identify the cars, motor cycles, buses, and trucks.

The other objective is to determine traffic density at various lanes of a traffic intersection. The system can also calculate the amount of vehicles in each lane by counting the cars that are captured in the lanes hence allowing the green light to be distributed to the lanes with the highest traffic density. Another purpose of the project is to install an emergency vehicles detection system, which will detect ambulances or other emergency vehicles in the traffic flow. In the event where an emergency vehicle is identified, the system will automatically give the lane priority by giving a green signal so the vehicle is not held up.

Besides, the project is expected to combine artificial intelligence algorithms with embedded hardware devices like Raspberry Pi, PIC microcontroller and traffic signal LEDs to establish a fully operational traffic signal automation system. The other project goal is to come up with an economical traffic management system that will not require the installation of expensive hardware sensors to detect vehicles but rely on camera technology to monitor the traffic. This will save on installation expenses and maintenance expenses and will give precise traffic monitoring capabilities.

3.4 Project Scope and Limitations

3.4.1 Scope of the Work

The field of the suggested AI based traffic lights automation system refers to the idea of enhancing the effectiveness of traffic lights at the road intersections by means of real-time car identification and smart signal management. The system will show how an artificial intelligence and embedded system can be combined to come up with smart traffic management systems.

Within the project, a Pi Camera is used in conjunction with a Raspberry Pi in order

to execute traffic monitoring. The camera records live photos of the intersection traffic and uses the YOLO object detector algorithm to detect cars and approximate the number of cars in the traffic. Depending on the traffic density observed, the Raspberry Pi transmits control messages to a PIC microcontroller which is used to control the traffic lights. The microcontroller turns the traffic lights based on the commands being sent by the Raspberry Pi.

There is also inbuilt system of a stepper motor mechanism that enables the camera to turn around and be able to watch the traffic in various directions. The system has this feature, which allows it to monitor traffic conditions at various lanes in the intersection. Though the given system is used as a prototype with demonstration purposes, its main idea can be applied to the real-life traffic intersections. The proposed solution can be used to bring smart transportation infrastructure to cities with additional upgrades and connection to the city-wide traffic management.

The system can be further improved in the future through adding cloud-based traffic monitoring systems, incorporating predictive traffic monitoring with machine learning models, and increasing the system to handle several intersections at once.

3.4.2 Limitation

Regardless of the good outcomes and the potential of the proposed intelligent traffic control system, some limitations should be mentioned. These limitations are mainly due to the environmental, hardware performance, data availability, and implementation scope. The knowledge of these limitations is useful in determining areas of development and inform future developments on large scale deployment. The existing use of the system has been tried majorly under a controlled or simulated setting. The arrangement is typically limited to small scale intersection, or laboratory conditions with a minimum amount of disturbance. Controlled setups allow the traffic movement, illumination and the conduct of the motor vehicles to be predictable and easily controlled.

But in real world intersections of city streets are much more difficult. These include multiple lanes, different types of vehicles, changing lighting and other people and vehicles that are not predictable. Assertions like system violation by traffic regulations,

crossing of pedestrians in unauthorized points, and unanticipated impediments, to name just a few, may influence system accuracy. Moving away the controlled to the live intersection of the city would need further calibration, adjustment of the cameras placement and more high-end computing to accommodate real-time and big data.

Additionally, an actual intersection has irregular characteristics like accidents, construction, or sudden traffic jams, all of which can hardly be modeled. These dynamics might involve the need to add layers of AI to detect anomalies and provide a decision, which are not in the scope at this point. Therefore, the prototype proves its functional efficiency, but its work in full-scale application in urban areas is still to be properly tested.

Lighting and Weather Dependence

The reliance on the environmental conditions, especially on the light and weather, can be taken as one of the biggest problems of any vision-based system. The quality of images obtained through camera is very crucial in detecting vehicles. The visibility of vehicles as well as pedestrians is significantly reduced during low-light conditions like at night time or when one is in the fog. It may lead to misclassification of the object or under detection of the object by the YOLO algorithm.

Likewise, unfavorable weather conditions such as excessive rain, fog and dust storms will warp or block the camera feed and they can provide inaccurate input to the image processing. An example is that the camera lens may experience water droplets or glare of streetlights that lead to the intrusion and noisy nature of captured photos. Even high-level image preprocessing filters, in this case, may fail to fully offset losses to visibility.

In order to reduce such concerns, the infrared cameras or thermal imaging sensors might be considered in the next round. Also the adaptive exposure control, real time brightness fix, and weatherproof camera covers can be used to ensure consistency. These improvements, however, can be costly to maintain the system and complicate the system.

Dataset Constraints

The other weakness is the dataset on which the AI model was trained. Accuracy of the YOLO

algorithm directly relies on the quality and the variety of its training. Without the types of vehicles that are present in the area, even if that area has auto-rickshaws, two-wheelers, and bullock carts, or even special emergency vehicles, the model might fail to identify them properly. It is especially important in such countries as India, where the combination of vehicles and traffic patterns highly differ with those of the datasets obtained in the Western countries. Moreover, the data sequence could be too distant to portray the variety of environmental differences, e.g. lighting conditions, camera distance, and obstructions (another car covered in the frame). These would to a margin decrease the accuracy of detection in the actual deployment. Customization of the dataset to region specific vehicle images, multiple environmental conditions and traffic jams may assist in the improvement of model robustness.

Scalability Issues

Although the suggested system works remarkably well with one intersection level, there are various challenges when applied to an urban city-wide network. Massive implementation would involve the coordinated communication systems at the intersections, to facilitate coordinated signals operation. Lack of proper communication would see a junction giving out a green signal leading to congestion in another locality. Real time communication among these intersections would need a good network connectivity probably 4G/5G or dedicated IoT networks like the MQTT. Moreover, the system would have to support very large data streams of multicast camera feeds across various camera streams at a time which may exceed bandwidth and storage capacity. To overcome them, distributed data processing and storage using cloud integration and edge computing may be adopted. Nevertheless, this also creates the further problems of data security, latency, and maintenance.

Besides that, there may be a need to have a local backup system in every intersection that can be used in case of temporary network failure. Further scaling can be achieved through a hybrid model in the expansion of these features, which may allow taking control on the field, but in the long term, scalability can be optimized with cloud-based analysis, which can be performed centrally. With all these possible solutions, a large infrastructure would still be needed to implement the solution full.

CHAPTER 4

PROPOSED SYSTEM

4.1 Introduction

Traffic congestion is one of the issues that are of great concern in the contemporary cities owing to the sudden growth of the number of vehicles. The traditional traffic signal systems work based on a timing system and are not concerned with the real time traffic situation at crossways. Consequently, such systems tend to produce unwarranted waiting time of cars and wastage of traffic.

In the traditional traffic controllers, the flow switches on the traffic lights at a set time interval irrespective of the car traffic present in the lane. This causes scenarios whereby less vehicle lanes are accorded the same signal time with heavy traffic jams lanes. This inefficient distribution of signals is a cause of traffic jams, high fuel wastage and pollution.

In order to beat these shortcomings, there is need to have intelligent traffic management systems, which are able to make real-time analysis of the traffic conditions besides dynamically manipulating the signal timings. The latest developments in the field of artificial intelligence and computer vision technologies allow building smart systems of traffic control, which are able to identify vehicles and determine the traffic density automatically.

The offered system states an automation system of traffic lights based on AI, which involves image processing and deep learning to track the level of traffic and manage traffic lights respectively. The system has used the Raspberry Pi Zero 2W as the central processing unit and Pi Camera to take images of real time traffic. The results of the captured images are processed by the YOLO (You Only Look Once) object detector algorithm that recognizes various kinds of vehicles (cars, motorcycles, buses, and trucks). The system, depending on the number of vehicles that are detected in the road, determines the traffic density and opens and closes traffic lights to help improve traffic flow.

Also, the system can detect emergency vehicles including ambulances and fire trucks. Upon detection of an emergency vehicle the system will automatically give the lane priority which will likely be given a green signal to enable the vehicle to go through the intersection in a hurry. This aspect aids in enhancing the speed of emergency response and increases road safety. The proposed system offers an intelligent, yet effective traffic management system that can be adjusted to the current traffic situation in real-time through the combination of artificial intelligence and embedded hardware units.

4.2 Introduction to the Proposed System

The suggested AI-based traffic light automation system is divided into hardware and software that has to interact to ensure the monitoring of traffic and the automatic control of traffic lights. The system combines computer vision technology and an embedded hardware to develop a smart traffic control system. The key element of the system is the Raspberry Pi Zero 2W that acts as the primary processing unit. The Raspberry Pi captures traffic images with the help of a Pi Camera and processes them with the help of the YOLO object detection algorithm. The algorithm identifies the position of the vehicles in the frames captured and identifies the vehicles.

Once vehicles are detected, the system would count the number of vehicles in each lane and determine the density of the traffic. The density values are calculated in order to decide which lane is supposed to be given priority. The green signal is given longer time to lanes with high vehicle density in order to get the vehicles to cross the intersection successfully. The signal control decisions are communicated to a PIC microcontroller through serial communication by the Raspberry Pi. The PIC microcontroller is the traffic signal controller which switches the traffic signal LEDs based on the commands it gets via the Raspberry Pi. The LEDs are traffic lights that are used to model the real-life traffic lights that are red, yellow, and green lights depending on the direction. These LEDs illustrate how the system regulates traffic in an intersection.

It also has a stepper motor which enables the camera to turn around and scan traffic in different directions. The stepper motor will be run by a ULN2803 driver IC that supplies the required current in the running of the motor. Regulated power supply ensures

that the system has a consistent voltage across all the electronic parts hence ensuring that the system operates reliably. The proposed system will integrate these hardware and software systems to form an intelligent traffic management system that can dynamically manage traffic lights in accordance to traffic conditions in real time.

4.3 Elements of the Proposed System

The proposed system will have a number of hardware and software units that can be used to achieve intelligent traffic signal control. The central processing unit of the system is the Raspberry Pi Zero 2W. It does image processing and runs the YOLO object detector algorithm that helps to recognize vehicles in the traffic flow. The OpenCV libraries are also included in the Raspberry Pi, which runs the Python programming language and image processing operations.

Pi camera module is employed to take the real-time picture of the traffic at the intersection. Video frames are constantly captured to the camera and sent to Raspberry Pi to be analyzed. Images that are captured give the required input data that is used in the vehicle detection.

The PIC microcontroller is involved in the control of the traffic light LEDs. The Raspberry Pi transmits control commands to it and the corresponding traffic lights are activated depending on the road conditions. The microcontroller will make sure there is proper switching between the red, yellow and green lights.

Traffic lights are the lights utilized in controlling the movement of vehicles and these are known as traffic signal LEDs. These LEDs represent the actions of real traffic lights and enable the system to show the smart control of the traffic lights. The camera is rotated using a stepper motor in order to monitor the traffic conditions in a variety of directions. This enables the system to collect traffic data on various lanes of the intersection.

The stepper motor is driven using the ULN2803 driver IC. It serves as an interface with the motor, and it gives the necessary current amplification to the motor, and the microcontroller. Controlled power supply will make sure that every component is fed with a stable voltage to operate adequately. Other elements like resistors, capacitors,

crystal oscillators assist in the operation of the microcontroller and other electronic circuits. A combination of these elements constitutes a fully established intelligent traffic monitoring and control system.

4.4 Workflow of the Proposed System

The functionality of the proposed system starts with the Pi Camera which is going to capture real-time traffic images at the intersection. The camera captures video frames and sends them to the Raspberry Pi which processes the video frames. Raspberry Pi is the device that processes every frame captured with the help of the YOLO object detection algorithm. This algorithm provides an analysis of the image and recognizes vehicles in the traffic stream. Around the detected vehicles, bounding boxes are created and a vehicle classified into a category of cars, buses, trucks, or motorcycles.

After the detection of the vehicles, the system will count the vehicles per lane. Traffic density levels are estimated with the number of identified vehicles. The system then correlates the levels of density of various lanes to ascertain the lane that needs priority. The system is dynamically adjusted in the timings of the traffic signals, based on the calculated traffic density. Lanes that have increased vehicle density are given a longer duration of green lights so that the vehicles can be able to pass through the intersection at a better rate. When an emergency vehicle like an ambulance is detected by the system, the system gives it priority by giving it a green signal. This enables the emergency vehicle to pass through the intersection within a short time without having to wait on the regular signal cycle.

Once the PIC microcontroller is identified with the correct signal control action, the Raspberry Pi transmits the commands to the PIC microcontroller via Raspberry Pi-PIC serial communication. These instructions are sent to the microcontroller and the relevant traffic light LEDs are triggered. The stepper motor has the ability of spinning the camera every now and then in order to monitor traffic conditions in the various directions. The ability to monitor and adjust to changing traffic situations keeps this system in an effective traffic flow.

4.5 Benefits of the proposed System

The proposed traffic light automation system that uses AI has a number of benefits compared to the traditional traffic signal systems. Among the initial benefits, there is the possibility to change the signal timings dynamically according to the real-time traffic conditions. This saves wastage of time and enhances efficiency in traffic. The other benefit is the vehicle detection and classification using artificial intelligence. YOLO object detection algorithm is accurate in its detection and allows the system to track various classes of vehicles in real time.

Another feature of the system is the emergency vehicle prioritisation which is significant to the current traffic management systems. The system allows the ambulances and other emergency vehicles to transit the intersections at a faster speed, thereby boosting the response time to an emergency and improving the safety of the people. Traffic monitoring with the help of camera allows avoiding the use of costly road surface hardware sensors. This saves a lot of installation and maintenance expenditure.

The suggested system can also be expanded and scaled to address various points of intersection in a city. As it continues to evolve, the system will be incorporated into the infrastructure of a smart city to facilitate solutions related to a high level of traffic control. All in all, the suggested system will reveal how artificial intelligence, computer vision, and embedded systems could be integrated to produce an intelligent traffic control platform that will help to enhance traffic movement and decrease congestion.

4.6 System Design Methodology

The proposed AI-based traffic light automation system has been designed on the basis of a structured approach which incorporates artificial intelligence, computer vision, and embedded systems to provide intelligent traffic signal control. The primary purpose of the system design is to watch traffic situations and make timings of signals dynamically depending on the traffic density.

The system design proceeds to the choice of the appropriate hardware parts including Raspberry Pi, Pi Camera, PIC microcontroller, and traffic signal LEDs. The Raspberry Pi has been selected as the main processing unit due to its option of executing

high level programming languages and the availability of libraries to process images needed to implement deep learning algorithms.

The second process of the system design is to capture images of traffic with the Pi Camera module. The camera is set in such a way that it is able to take clean shots of the traffic coming into the intersection. Such images are sent to Raspberry Pi to proceed with the processing.

Once the images are captured the system identifies the vehicles in the traffic stream by using the YOLO object detection algorithm. Each frame of the video captured is fed into the algorithm which identifies vehicles in the form of bounding boxes. The vehicles detected are then counted in order to identify the density of traffic in each of the lanes. Depending on the values of the calculated traffic density, the system defines which one is supposed to be given priority and changes the signal timings.

The signal control commands are then sent out of the Raspberry Pi to the PIC microcontroller which drives the LEDs of the traffic lights. The microcontroller implies the proper traffic lights to control the traffic at the intersection. The whole system works in a continuous mode, which enables it to keep up with the changing traffic conditions and efficient flow of traffic.

4.7 Data Flow of the Proposed System

The information flow of the proposed AI-based traffic lights automation system is sequential entailing traffic surveillance and culminating with traffic control. Traffic analysis requires one to understand the flow of data on how the system processes the traffic data and the decision making process. It takes a Pi Camera that records the real time image of the cars coming to the intersection. The operations of the camera are transferred to the Raspberry Pi via the interface of the camera.

After receiving the images through the Raspberry Pi, the images are processed with the help of the YOLO object detection algorithm. The algorithm processes the picture and identifies cars in the frame. A bounding box and classification information is used to represent each identified vehicle. Once the vehicles have been identified, the system counts the amount of vehicles that are in each lane. Traffic density is estimated with this

information, and the lane that is to be given priority is determined.

According to the computed values of density, the Raspberry Pi produces signal control commands. Serial communication of these commands is sent to the PIC microcontroller. The commands are sent to the PIC microcontroller which turns on the respective traffic light LEDs. Depending on the conditions of the traffic, the LEDs show red, yellow, or green lights. This stream of data allows the system to know the traffic profile on the road and make traffic lights dynamic.

4.8 Priority of Emergency Vehicles

Ambulances, fire engines, and police cars are emergency vehicles that must have fast access at the traffic intersections so that they can provide emergency response. Traffic light delays can also impact the emergency response time greatly and can also be a cause of severe repercussions.

The suggested system also has the emergency vehicle priority system which identifies the emergency vehicles through the YOLO object detection algorithm. This algorithm is trained to identify certain kind of vehicles such as ambulances and other emergency service vehicles. Once an emergency vehicle is detected within the traffic stream, the system automatically interrupts the regular traffic signal cycle and gives the traffic in that lane a green light. This enables the emergency car to go through the intersection immediately.

After the system clears the intersection the emergency vehicle passes, the system goes back to normal operation of the traffic lights depending on the density of traffic. This will make sure that the emergency vehicles are given priority without affecting the efficient flow of other vehicles. The additional feature of the traffic management system, which is the integration of emergency vehicle detection, plays a significant role in the functionality of the system and the improvement of the safety of the population. The offered AI-powered system of traffic lights automation can be implemented under different transportation and urban settings. Among the major areas of application of the system are in traffic intersections of urban areas where traffic jam is a major issue.

The system can be connected with centralized traffic management platforms in the

smart city setting to track and regulate various intersections at the same time. This enables the traffic authorities to study the traffic patterns and determine the timing of the signals to the point of maximum optimization to the entire transportation network.

The system is also applicable on highways and road junctions whereby traffic is very high during different times of the day. The system will be able to assist in minimizing the congestion and enhancing the traveling efficiency by dynamically changing the signal timings.

The second significant use of the system is managing emergencies. The system can also be used to reduce the time taken by the emergency services to respond to emergencies and make the emergency services more effective by giving the ambulance and other emergency vehicles priority clearance. The system can also be applied to research and learning institutions to show how artificial intelligence, computer vision and embedded systems can be integrated in intelligent transportation systems.

CHAPTER 5

SYSTEM ARCHITECTURE

5.1 Introduction

System architecture explains the general structure and the operational model of the planned AI-based traffic light automation system. It describes the interaction of different hardware and software elements that are used to monitor intelligent traffic and automatically control signals. An effective architecture is used in the interpretation of the way the system gathers data, processes information and generates control outputs that control the flow of traffic at intersections.

The proposed system architecture will incorporate the use of computer vision methods and embedded hardware to form a smart traffic system. The conventional traffic systems use timetables that are fixed and not depending on the real-life traffic. Conversely, the suggested system is constantly checking the traffic density through cameras and artificial intelligence algorithms that so that the signal timings can be varied.

The main element of the architecture is the Raspberry Pi Zero 2W which serves as the central processing unit. The Pi Camera module provides the Raspberry Pi with the video input and processes the images using the deep learning algorithms like YOLO (You Only Look Once). These algorithms identify traffic and find out the density of traffic in real time. According to the data analyzed, the system produces control signals in order to control traffic lights.

It also has a system architecture that incorporates a PIC microcontroller with which it acts as the traffic signal controller. Raspberry Pi sends the commands to the microcontroller which in turn triggers the traffic lights. Also a stepper motor mechanism enables the camera to swing and survey the traffic situations in various directions. The given architecture will allow the system to keep running and adjust to real-time traffic changes. The proposed architecture can be used to develop smart transportation solutions in the current urban settings by combining artificial intelligence with embedded hardware platforms to offer a versatile and efficient solution.

Moreover, the architecture will be designed to allow the development of the system in a modular way, that is, every functional unit of the system will be able to work on its own, and yet, contribute to the functioning of the traffic management system. It is a modular method of enhancing system reliability and can be upgraded in the future without the need to entirely redesign the system.

The flow of information that occurs between the input devices, processing units and control modules in the system is another significant part of the system architecture. Pi Camera is the input device which receives the visual information of the traffic world. The analysis of these visual data is carried out using the Raspberry Pi that detects and processes the movement of vehicles and calculates the level of traffic jam. The resulting processed data is then transferred into control commands that control the operation of traffic lights using PIC microcontroller.

The architecture proposed also provides effective interaction of software algorithms and the physical hardware components. The software component can be involved in the complex activities like image analysis, vehicle recognition and decision making whereas the hardware components implement the decisions by controlling the traffic signal lights. This is a synchronized communication between software and hardware that makes the system able to react fast to the changes in traffic conditions.

The other benefit of the proposed architecture is that it is scalable. The system would be able to accommodate a number of cameras and intersections to enable the traffic monitoring system to be used to cover wider areas of the city. The system can help the formation of intelligent transportation infrastructure in smart cities by incorporating several units into one centralized traffic control system. Moreover, the architecture facilitates real time monitoring and automatic control and eliminates manual operation of traffic signals. The system allows traffic authorities to access the system remotely and study the pattern of traffic to improve more on traffic management strategies.

In general, the system architecture offers a detailed system that incorporates computer vision, artificial intelligence, and embedded system technologies in development of a smart traffic signal automation system. The proposed structure will be

designed to enhance the efficiency of traffic flow, minimize the traffic congestion, and increase the safety of roads through the dynamism of the traffic lights in accordance with the real-time traffic conditions.

5.2 Block Diagram

The block diagram in Fig 5.1 of the proposed AI-based traffic light automation system is a functional relationship between the various hardware and software components employed to achieve intelligent traffic signal control. It gives a visual image to illustrate the flow of data within the system and the role of each component in monitoring the conditions of traffic as well as regulating the traffic lights. The block diagram is also useful in knowing the overall functioning of the system, how traffic images are captured to the management of the traffic lights depending on the traffic density.

The system consists of a number of interrelated modules which carry out particular functions like traffic monitoring, image acquisition, vehicle detection, traffic density analysis, decision making and traffic signal control. All the modules interact with the other modules to bring about smooth and efficient running of the system.

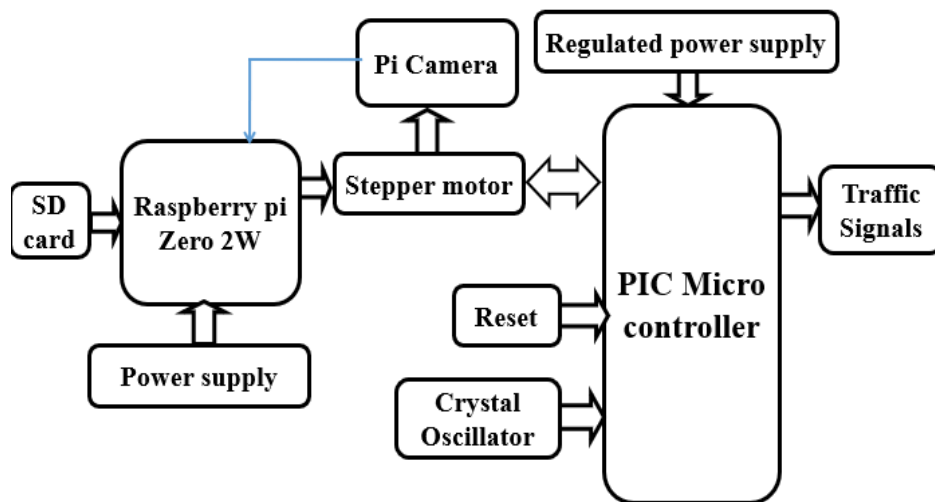


Fig .5.1 Block Diagram

The Pi Camera module is the first block in the system architecture, which is the input device of the entire system. The Pi Camera is installed in a manner that it captures the clear images of the approaching vehicles to the traffic intersection. The camera is used to

record video frames and take real-time traffic pictures. These pictures give the raw information on which to analyse the state of traffic. The captured frames are sent directly to the Raspberry Pi to be further processed using the Raspberry Pi. By utilizing camera-based monitoring system, the expensive road sensors are not necessary and the system can also be used to view the conditions of the traffic.

The key to the system is the Raspberry Pi Zero 2W that is used as the central processing unit of the system and does all the significant computational processes. It takes the image data of the Pi Camera and processes the frames that it captures with sophisticated image processing and deep learning algorithms. The Python-based program that runs on the Raspberry Pi incorporates the libraries of computer vision including OpenCV and the YOLO object detection model.

Identification of vehicles in the traffic images captured is done by the YOLO (You Only Look Once) algorithm. YOLO is an object detection algorithm, which is real-time and which divides an image into grids and determines objects in each cell of the grid. It creates bounding boxes of every recognized cars and classifies it into various categories, including cars, motorcycles, buses, and trucks. The algorithm is also very fast in this process of detection and therefore can be used in real time in relation to traffic monitoring applications.

When the vehicles have been detected, the Raspberry Pi compares the detection results and the number of vehicles in each of the lanes is determined. Information on the number of vehicles is applied to determine the level of traffic congestion at the intersection. Traffic density is a significant parameter that dictates the traffic control of the traffic lights. In case the vehicle traffic on one lane is more than in other lanes, the system gives this lane a longer green signal. The Raspberry Pi carries out decision-making after having the values of density calculated. It also compares the density of various lanes and selects which lane is supposed to be prioritized. The system then makes up control commands using defined traffic control logic. Such directives determine the traffic light that needs to be green and the duration of time the signal is required to be active.

After the decision on the signal control has been made, the Raspberry Pi transmits control signals to the PIC microcontroller via serial communication. Serial

communication is a sure way of relaying data between the microcontroller and the Raspberry Pi. The instructions transmitted by the Raspberry Pi command the microcontroller on the operating method of the traffic lights.

The microcontroller is the PIC microcontroller, which acts as the traffic signal controller. It is controlled by the Raspberry Pi and is able to act on its instructions by turning the respective traffic light LEDs on or off. The micro controller will switch the red, yellow and green lights as per the instructions given by the processing unit. A microcontroller is used to correct the timing and ensure the traffic lights can be controlled properly.

The system uses traffic signal LEDs to replicate the real world traffic lights. These LEDs are the three primary states of the traffic light namely red, yellow and green which indicate stop, caution and go respectively. The LEDs are positioned in such a way that they depict various directions of traffic flow where the intersection takes place. The system shows the manner in which real traffic lights work based on the conditions on the road by regulating these LEDs.

The other valuable part of the system is the stepper motor mechanism which is made to move the camera. The conditions of the traffic in the various directions of the intersection can be different and by monitoring only one direction the traffic might not give all the information. To circumvent this shortcoming, the stepper motor turns the camera at regular intervals that will enable it to take pictures in different directions. The rotation is what enables the monitoring of the traffic conditions of various lanes in the intersection.

The ULN2803 driver IC is used to control the stepper motor. The driver IC provides a connection between the motor and the microcontroller. The high current needed to operate the motor cannot be supplied by the microcontroller directly and therefore, the driver circuit increases the size of the control signals and delivers the current needed to move the motor. This provides high efficiency of the motor.

The whole system is in the continuous mode which means that the system is able to observe the traffic conditions in real time. The images are captured with the camera,

processed with the Raspberry Pi with the help of artificial intelligence algorithms, and the microcontroller regulates the traffic lights. The ongoing process of monitoring, processing and controlling of the signal makes the system adapt dynamically to the changing traffic conditions.

On the whole, the interaction of various elements of the system is evident in the block diagram. It demonstrates the process of visual traffic data gathered by the camera to process it with the help of artificial intelligence algorithms and translate it into the control signals that will manage the traffic flow. The architecture is the base of the intelligent automation system of traffic lights and allows managing traffic at crossroads effectively.

5.3 Functional Modules of the System

The system that is proposed as an AI-based traffic light automation is developed with a modular architecture where the entire system is subdivided into several functional modules. The modules have a defined operation in which they aid in intelligent tracking and management of traffic lights. The modular design also enhances the efficiency, scalability and maintainability of the system since every module can be coded, tested and even enhanced without any impact on the overall functionality of the system. The system is sequential in all the modules. Traffic monitoring has to start followed by the image processing, vehicle detection, traffic density estimation, decision making and lastly traffic signal control. These modules are synchronized to maintain traffic conditions are under constant surveillance and traffic lights adjusted dynamically depending on the real-time conditions.

The traffic monitoring module is the initial module in the system. The major component of this module is the Pi Camera which is the visual sensing unit of the system. Camera has been placed at the cross of the traffic intersection in such a way that it could see the approaching vehicles in various directions clearly. Pi Camera is constantly taking video frames and real-time images of the traffic environment. These are the captured frames which are the main input data to the intelligent traffic monitoring system. The camera is continuously capturing the activity of the traffic and as such, the system will be able to monitor the alteration in flow of traffic and dynamically analyze traffic density.

The second one is the image processing module. This module is executed with the help of the Raspberry Pi Zero 2W and it has the task of processing images taken by the camera. The processing of the image is used to enhance the quality of the acquired frames and to prepare them to be analyzed. Image processing libraries like OpenCV are used in the Raspberry Pi to carry out functions like resizing of images, eliminating noise, extracting frames and enhancing contrast in images. These preprocessing functions are used to make sure that the images are clear and ready to be used in detecting objects. The image processing module is also important in enhancing the accuracy of vehicle detection because it prepares the captured frames to be analyzed further.

The third module is the vehicle detecting module. The module applies the YOLO (You Only Look Once) deep learning algorithm in order to identify vehicles in the traffic photo taken. YOLO is a sophisticated object detector which can detect several objects in one frame with a high level of accuracy and speed. The algorithm processes the individual frame and creates bounding boxes on cars identified. Cars, motorcycles, buses, trucks or emergency vehicles are other categories, which each of the detected objects is categorized into. YOLO is very appropriate in the intelligent traffic monitoring systems due to the capability of identifying several vehicles at a time. This module is the main intelligence of the system as it enables the system to be aware of the happenings in the traffic environment.

The fourth module is the traffic density estimation module. After measuring the vehicles with the help of the YOLO algorithm, the system determines the number of vehicles in each lane of the intersection. This fact is utilized to obtain an estimation of the level of traffic density in every direction. The estimation of traffic density is a significant procedure since it defines the manner in which traffic signals would be managed. When a specific lane has many vehicles, the lane will be said to have high vehicle density. Conversely, lanes that have fewer vehicles are said to have a low level of traffic density. The system will be able to detect and prioritize the lane with the highest density by comparing the density of various lanes.

The fifth module is the decision-making module. The information on the density of the traffic calculated in this module is used to decide how traffic lights are to be

administered. The Raspberry Pi receives the density values and uses logic which has been set to define the best signal timing. To illustrate, when there is a huge traffic congestion in one lane relative to other lanes, the system will allow that lane to take up a longer green signal. The decision-making module is used to make sure that the traffic lights are dynamically operated instead of adhering to time schedules. This signal control system is adaptive and enhances traffic, as well as minimizes the time spent at a crossing.

The last module is the traffic signal control module. This module has the role of implementing the decisions of the processing unit. It is comprised of PIC microcontroller and traffic signal lights. The PIC microcontroller receives control messages by the Raspberry Pi via an internal serial connection. These commands are sent to the microcontroller and the relevant traffic lights are switched on. The red, yellow and green lights of the intersection are depicted by the traffic signal LEDs. The microcontroller makes sure that the signals switch accordingly in the order that they are supposed to and that they are on at the required length of time.

Besides the control of the traffic lights, the microcontroller can also be used to control the stepper motor that turns the camera. The rotation mechanism enables the system to check the traffic condition of various directions of the intersection. The system will also be able to view traffic on various lanes by rotating the camera on a regular basis to enhance the quality of traffic monitoring.

These functional modules are cyclic and they are in a constant process. The Raspberry Pi processes the images, which are taken by the camera, recognizes vehicles, and the system calculates the density of traffic, and the microcontroller operates the traffic lights respectively. This is an endless loop that enables the system to be dynamically responsive to traffic conditions. All in all, the system has a modular structure that guarantees the efficiency of work, the easy scalability of the system, and the enhanced stability. All these modules make the traffic control system intelligent as it is able to check the current state of traffic in real time and dynamically adjust the times of the signals to ensure that they minimize the number of congestions and enhance the movement of traffic.

5.4 Hardware and Software integration

The automation system of traffic lights is proposed as a combination of hardware and software development to allow building a smart and automated traffic control system with AI. The hardware components will do the sensing of the environment and control action, whereas the software components will do data processing, analysis and decision making. These two components together allow the system to monitor traffic conditions dynamically and in real time and automatically regulate the traffic lights.

The software aspect of the system is very important in the analysis of the traffic data recorded and the smart decision making on the basis of the recorded traffic situation. Raspberry Pi Zero 2W is the primary computing device that executes Python-based code regarding picture processing and automobile recognition. Python is chosen as a programming language due to its wide range of machine learning and computer vision libraries.

The vehicle detection uses the YOLO (You Only Look Once) deep learning algorithm to implement the system. YOLO is very popular in computer vision systems because it can recognize various objects in one image in real time. The video frames taken by the Pi Camera are processed by the algorithm that determines the presence of vehicles in the traffic scene. It creates bounding boxes of identified vehicles and categorizes them into various subgroups including cars, motorcycles, buses, and trucks. The system can use the results of these detections to estimate the density of traffic and the number of vehicles in each of the lanes.

The process of detection is supported with the help of image processing libraries (OpenCV). OpenCV offers the functions that are used to process image data such as frame extraction, resizing, filtering, and enhancement. These processes enhance the quality of the images that are captured and get them ready to detect the objects accurately. OpenCV coupled with the YOLO model allows the detection of vehicles in real time in an efficient and reliable manner.

The hardware part of the system consists of a number of devices that co-exist to execute the traffic control mechanism. A Pi Camera module is employed to get real-time images of drivers coming to the traffic intersection. The camera is directly attached to the Raspberry Pi and feeds off to stream video frames to be processed by it.

The traffic signal controller is PIC microcontroller. Although the Raspberry Pi can be used to perform multi-faceted computational functions, including image processing and decision making, the microcontroller manages the low-level control functions needed to switch traffic lights. This division of labor enhances efficiency and dependability of the system.

Traffic signal LEDs refer to the LED symbols which are linked with the PIC microcontroller and are the red, yellow and green lights used in the traffic management in the real world. These LEDs replicate the work of the traffic lights and show how the movement of vehicles is controlled at a crossroad. The microcontroller is used to turn on the right LEDs depending on the commands sent over by the Raspberry Pi.

Serial communication protocols are used to create a communication link between the Raspberry Pi and the PIC microcontroller. Serial communication offers a sure way of communicating with the devices. The Raspberry Pi sends control signals to the microcontroller via the serial interface when it is done processing the traffic images and determining the right time to send a signal. These commands are received by the microcontroller and the respective signal control actions are implemented.

A stepper motor is another significant equipment in the system and it is used to spin the camera. Intersections are usually targeted by more than one lane and direction and the surveillance of only one direction would not give a full information about the traffic. The system will be able to provide images in other angles by turning the camera with the help of a stepper motor and allow a full view of the traffic setup.

The ULN2803 driver IC is used to control the stepper motor. The motor and the microcontroller are connected via the driver IC. As microcontroller is not capable of delivering the large current, needed by the motor, the driver circuit boosts the control signals and delivers enough current to operate the motor. This guarantees steady and

unhindered movement of the camera.

The combination of both software and hardware units forms a closed system where traffic information is recorded, processed and applied to the traffic lights constantly. The camera gathers visual information, the Raspberry Pi handles the information through artificial intelligence algorithms, and the microcontroller handles the control instructions through running the traffic lights.

This integrated system which is smooth between hardware and software enables quick response to change in traffic conditions. When the traffic density in a given lane is more than average then the system automatically assigns a longer green signal in the lane. Likewise, in case of low traffic density, the time frames of the signals are also changed to ensure the traffic moves efficiently.

Besides the benefits of increased efficiency of the traffic, the combination of hardware and software allows the system to accommodate other functions like emergency vehicle identification and camera rotation. These characteristics increase the functionality of the traffic management system and help develop the intelligent transportation systems. In general, the seamless interaction of hardware and software was the key to the effective work of the offered AI-based traffic light automation. It allows the system to have real time traffic monitoring, smart decision making and automated control of the signals, hence, making traffic flow, congestion, and road safety better.

5.5 Data Processing

The signals processing and data flow are also significant to the functionality of the proposed AI-driven traffic light automation system. This section describes the flow of information in the system i.e. the flow of information beginning with the traffic monitoring and finally the traffic signal control. The system works with visual huge amounts of data that are taken in the traffic environment and this data is transformed into control signals which control vehicle traffic at intersections.

The general flow of the data in the system starts with the Pi Camera component, which serves as the main input point in the system. The camera tracks actual time pictures

and video shots of moving cars and vehicles in the intersection. These frames are captured as it is a raw value of the data needed to analyze traffic. Traffic conditions are dynamic and as such, the camera takes about continuous captures several times in a given period of time so that the system is continuously supplied with updated information about traffic.

After the pictures are captured they are sent over to the Raspberry Pi Zero 2W where they are processed. Raspberry Pi is the main processing unit whereby images can undergo analysis and the car detected. Image processing techniques are first applied on the captured frames. Before carrying out approximation, the image can be preprocessed by resizing, brightness and contrast, and noise reduction. Through these operations, the quality of the image is improved and ready to detect the objects accurately.

Once the frames are pre-processed, the processed frames are forwarded to the YOLO object detection algorithm. YOLO is the deep learning model that is specifically created to detect objects in real-time. It examines the whole image in a single pass and finds objects appearing in the image. Within the framework of the suggested system, the algorithm identifies vehicles, including cars, motorbikes, buses, and trucks. The algorithm creates bounding boxes on vehicles detected, and labels each of the detected objects.

The second stage in the process of data flow is the extraction of useful information in the results of the detection. The system identifies the count of vehicles that it has detected within a frame and classifies them based on their location within the picture. With the invention of the location of the identified vehicles, each vehicle is represented by the corresponding lane. The result of this information is calculated to get the traffic density in each lane of the intersection.

The estimation of traffic density is an important component of signal processing. The system calculates the amount of vehicles in each lane and the one having the greatest traffic density is denoted. When one of the lanes has considerably more traffic than the others, the lane is regarded as a high-priority lane. The signal timing necessary to clear up the traffic is made out of this information.

After calculating the traffic density, the information is processed by the decision-

making algorithm and the correct answer control commands are issued. The algorithm establishes the amount of time a traffic signal should be turned green and which signal to turn green. The logic is determined beforehand, with the lanes with the high vehicle density being prioritized by the decision. Once this process of making decisions is done, the Raspberry Pi will produce control signals and transmit them to the PIC microcontroller via serial communication. Serial communication interface also makes sure that there is reliability in data transfer between signal control module and processing unit.

The PIC microcontroller gets the control commands and processes the control command to control the traffic signal LEDs. Depending on instructions, the microcontroller will trigger the red, yellow or green lights respectively according to any direction of traffic. It is the last phase in the signal processing sequence in which decoded data is translated into real-life activities by which traffic flow is controlled.

The system also uses the signals that control the stepper motor and rotate the camera besides giving commands to traffic signals. The microcontroller provides an input to the ULN2803 driver IC which drives the stepper motor. The camera spin enables the system to take traffic images in varying directions of the intersection. This enhances better coverage of the monitoring system as well as the observation of traffic conditions of various lanes.

The whole process of data flow is continuous in circle. The camera takes new pictures, Raspberry Pi processes the pictures through the YOLO algorithm, traffic density is computed and the microcontroller reads the traffic lights respectively. This dynamism allows adaptation of the system to face the dynamic conditions of traffic.

The other feature of data flow system that should not be overlooked is that the system can synchronize itself to changes in traffic at a fast rate. The image processing and vehicle detection processes are done on a real time basis, meaning that the system will be able to identify the existence of a traffic jam and almost instantly alter the timing settings of the signals. This is because it allows the quick response system which can contribute to minimizing congestion in traffic and reinforce the overall effectiveness of

the traffic management.

The data flow architecture also provides structure such that every part of the system has a well-defined functionality. Data acquisition is done by the camera, data processing and analysis are done on the Raspberry Pi and the microcontroller conducts control actions. This division of labor also enhances reliability of the system and leads to an easier time of maintaining the system and upgrading in order to use it in the future. On balance, the intelligent traffic light automation system proposed is based on the data flow and signal processing mechanism. The system can dynamically regulate the traffic lights and enhance the movement of traffic in busy junctions by capturing, analyzing and responding to traffic data.

CHAPTER 6

HARDWARE COMPONENTS

6.1 Introduction

Hardware devices are important in the execution of the suggested AI-based traffic light control system. The functions of these parts include collecting the traffic information, processing signals and the management of the traffic lights according to the decision made by the software algorithms. The combination of appropriate hardware equipment allows realizing the ability of the system to work efficiently, and react promptly to the current traffic situation in real time.

The proposed system comprises a mix of embedded hardware functions and sensor devices to track traffic and regulate the functionality of signals. These hardware components are used with the software algorithms installed on the Raspberry Pi to form a complete automated system of managing traffic.

The hardware architecture primarily comprises of Raspberry Pi Zero 2W, Pi camera module, PIC microcontroller, ULN2803 driver IC, stepper motor, traffic signal LEDs, crystal oscillator, reset circuit and regulated Power supply. Each of these elements has a certain role in the system and helps to achieve the process of the entire intelligent traffic control mechanism work.

Raspberry Pi serves as a central processing unit that is in charge of running the image processing algorithms and identifying the vehicle with the help of the YOLO model. The Pi Camera records live images of the traffic which the Raspberry Pi process to detect traffic density. The PIC microcontroller will work as the traffic signal controller and will control the flow of lights according to the instructions of the Raspberry Pi. The stepper motor that drives the camera in order to monitor various directions of a traffic is controlled by the ULN2803 driver IC.

The traffic signal LEDs imitate the work of actual traffic lights through the

depiction of red, yellow, and green traffic signals in various directions. Crystal oscillators and regulated power supply circuits are other elements that guarantee stable and dependable functioning of the hardware system. Each hardwiring component that will be used in the proposed traffic signal automation system is explained in the sections that follow.

6.2 Raspberry Pi Zero 2W

The central processing unit in the proposed AI-based traffic lights automation system is the Raspberry Pi Zero 2W. It is a small inexpensive single-board computer that can be used in complex computational tasks (such as image processing and deep learning inference). The Raspberry Pi Zero 2W can be successfully used in embedded systems and IoT applications because of its small size, low power consumption, and enough processing capacity.

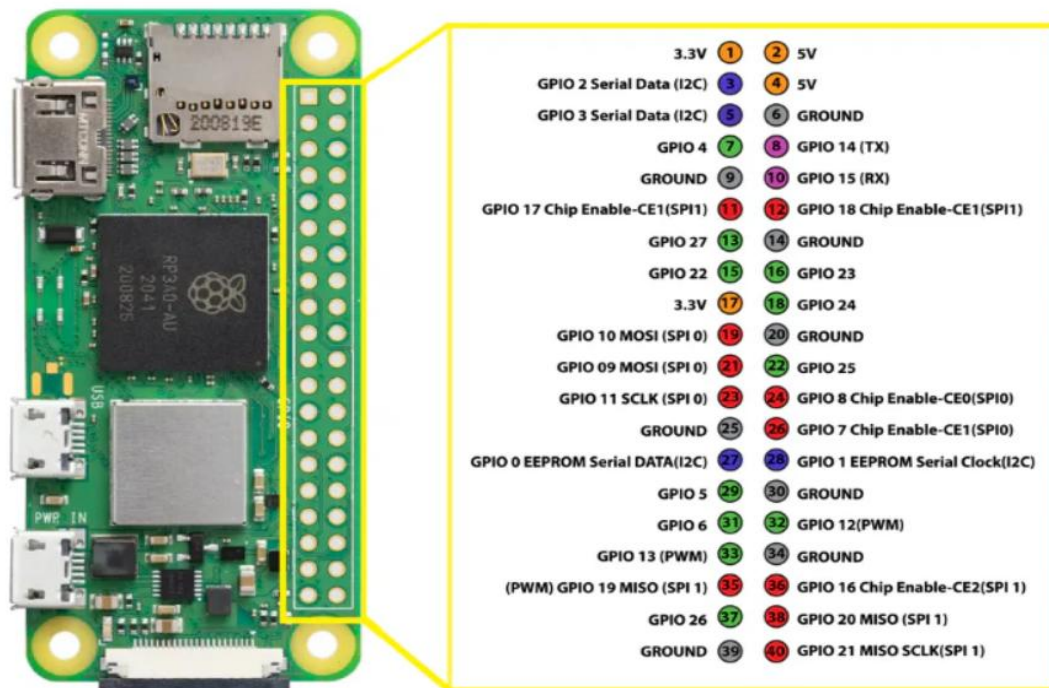


Fig .6.1 Raspberry Pi Zero 2W

The Raspberry Pi shown in Fig 6.1 is the suggested system will operate the Python-based traffic monitoring program and make use of the YOLO object detection algorithm. the Pi is connected to the Pi Camera module, which feeds the Pi with video sequences

and the processed images are used to detect the availability of vehicles in the traffic scene. Raspberry Pi relies on computer vision infrastructure like OpenCV to process the images obtained after which they are forwarded to the YOLO detection model. The YOLO model processes the images and recognizes the cars in the form of bounding boxes that are obtained around the objects. This system will then tally the vehicles registered per lane and then the density of traffic will be calculated.

According to the information on traffic density, the Raspberry Pi decides on the right time to set the traffic signal of each lane. After a decision is made the Raspberry Pi transmits control commands to the microcontroller in the form of serial communications to the PIC microcontroller. The microcontroller will subsequently switch on the relevant traffic lights.

The other benefit of Raspberry Pi in this system is that it is compatible with several hardware interfaces which include: GPIO pin, UART communication, camera interface and USB ports. These interfaces enable the Raspberry Pi to inter-communicate with other hardware components including microcontrollers, sensors, driver circuits among others. A powerful small board capable of being used in numerous other devices is the Raspberry Pi Zero 2W which runs on a quad-core ARM Cortex-A53 processor and has the capabilities of wireless communication through Wi-Fi and Bluetooth. The following capabilities render it appropriate in terms of executing advanced traffic monitoring systems and linking the system with future smart city infrastructure. All in all, Raspberry Plus 2W will form the brain of the proposed traffic management system and will carry out image processing, identify cars, calculate the traffic density, and make decisions to control the intelligent traffic lights.

6.3 Pi Camera Module

The Pi camera module shown in Fig 6.2 is a valuable hardware part in the proposed AI-based traffic lights automation system. It is the main sensory unit, which takes real-time images and videos of the intersection traffic. The input data to the image processing and vehicle detection algorithms are these images which are applied to the Raspberry Pi.

The Pi Camera itself is created to interface with the Raspberry Pi boards and directly

connect with the Raspberry Pi via the Camera Serial Interface (CSI) port. This interface enables high speed data transfer between the camera and the processing unit, thus the system is able to capture and process images in an efficient way.

The Pi Camera is used in the proposed system to capture video frames of the vehicles in the vicinity of the traffic junction. The camera is placed in a manner that there is a clear view of lanes in the traffic. The video frames that have been captured are further sent to the Raspberry Pi where they are processed and analyzed.

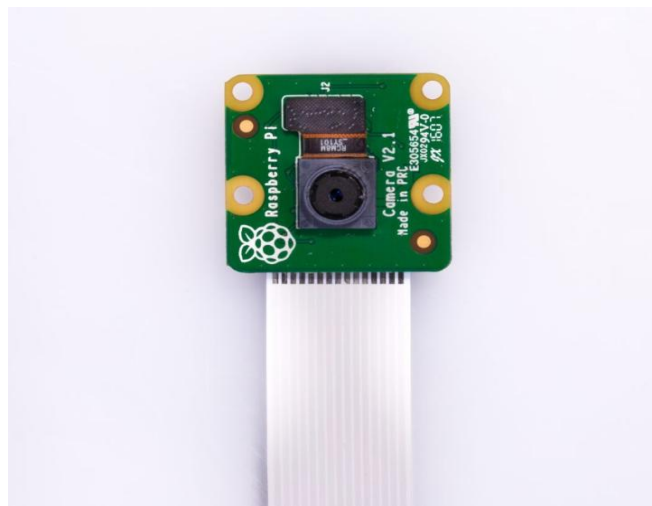


Fig .6.2 Pi Camera

The main objective of the Pi Camera in this system is to give visual information needed in detecting vehicles and estimating the density of traffic. The computer vision and the YOLO object detection algorithm are performed on the captured frames on the Raspberry Pi. The algorithm is used to analyze every frame and to find vehicles in the traffic scenario. Pi Camera module offers high resolution of image capture and it enhances the accuracy of vehicle detection. Good quality photos enable the deep learning algorithm to detect vehicles more accurately even in cases where more than one vehicle appears in the frame. This assists the system in approximating the traffic density better.

The other benefit of the Pi Camera is that of being small and consuming less energy. The camera module does not occupy a lot of space since it can be easily attached to the traffic poles or other structures near the intersection. It has a lightweight design, which makes it fit in embedded traffic monitoring systems. The camera is also compatible with

alternate frame rates and resolutions which means that the system is capable of taking images depending on the processing power of the Raspberry Pi. In real time traffic surveillance applications, the camera usually records frames at a middle frame rate to trade off processing rate and resolution.

Besides the vehicle detection, the camera may also be used to recognize emergency vehicles like the ambulances or fire tank when used with the proper detection algorithm implementation. This feature allows the system to give priority clearance to the emergency vehicles by dynamically changing the traffic lights. In general, the Pi Camera module is an essential element of the proposed traffic monitoring system as it gives the traffic monitoring the necessary visual input. It will allow the system to monitor traffic situations in real time and help to make intelligent decisions on traffic signals control.

6.4 PIC Microcontroller

One of the hardware elements that will be deployed in the proposed AI-based traffic light automation system is the PIC microcontroller. It serves as the traffic signal controller which controls the working of the traffic lights according to the instruction given by the Raspberry Pi. Embedded systems have microcontrollers as they can effectively carry out control functions at low energy consumption and high reliability.

The PIC microcontroller used in this project will manage the traffic signal LEDs which are used to depict the red, yellow, and green lights of various directions of traffic. The Raspberry Pi communicates with the microcontroller via serial communication and sends it commands on the type of traffic lights to turn on or off.

The PIC microcontroller applied in this system is normally the PIC16F877A, a commonly used microcontroller in embedded systems. It is founded on Harvard architecture and is meant to offer effective implementation of instructions at low power consumption. The microcontroller has a number of in-built peripherals in the form of input/output ports, timers, communication interfaces, and memory modules that can be used in real-time control applications.

The capability to drive several input and output devices may be considered one of

the major benefits of working with the PIC microcontroller. The microcontroller has a number of I/O ports, which may be set as either input or output pin. These I/O pins shown in Fig 6.3 are employed in the proposed system to control the traffic light LEDs as well as the stepper motor that will rotate the camera.

The Raspberry Pi transmits the instructions of traffic control to the microcontroller via a communication interface. After the instructions have been received, the microcontroller interprets them and switches-on the relevant traffic lights. An example is that as the Raspberry Pi detects that a given lane has more traffic it sends a signal to the microcontroller to switch the green light on that lane. The PIC microcontroller is also used to control the flow of traffic lights in addition to the stepper motor which rotates the camera. The stepper motor enables the camera to scan the traffic in various directions of the intersection. The motor is driven by a driver IC, the ULN2803 which increases the signals sent by the microcontroller.

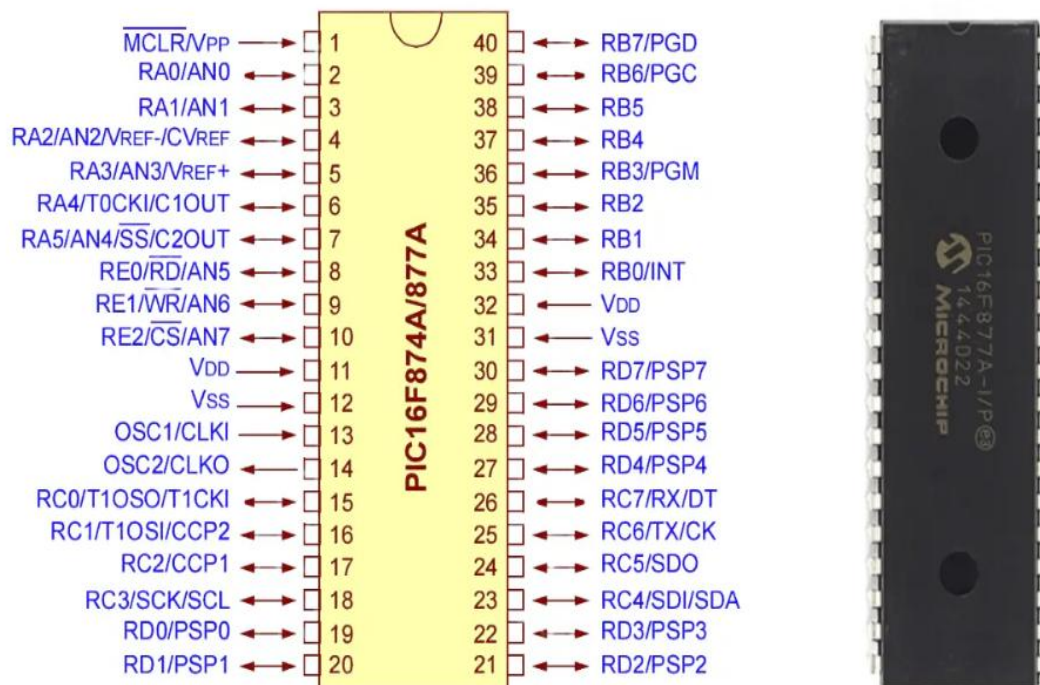


Fig .6.3 PIC Microcontroller

The PIC microcontroller is powered by a clock signal produced by crystal oscillator. The oscillator is used to give correct timing signals which coordinate the work of the

microcontroller. Timing is also crucial in terms of switching on and off of traffic lights correctly and ensuring correct signaling time.

The other notable aspect of the PIC microcontroller is that it can be used to execute embedded programmes in a language like the Embedded C. These programmes determine the way the microcontroller is expected to react to the instructions issued by the Raspberry Pi and act on the traffic lights.

The microcontroller also has the reset circuit in which it can restart safely when there are errors in the system or its power interruptions. This is to guarantee stability in the functioning of the traffic control system. On the whole, the PIC microcontroller is the hardware control unit of the proposed system. Although the Raspberry version can be used in complex computational processes like car image processing and vehicle recognition, the PIC microcontroller can be used in the implementation of the final control process because it controls the motor mechanisms and traffic lights.

The system takes the PIC microcontroller and the Raspberry Pi to provide a combination of powerful artificial intelligence processing and solid embedded control. This hybrid system enhances the efficiency, flexibility, and reliability of the intelligent traffic signal automation system.

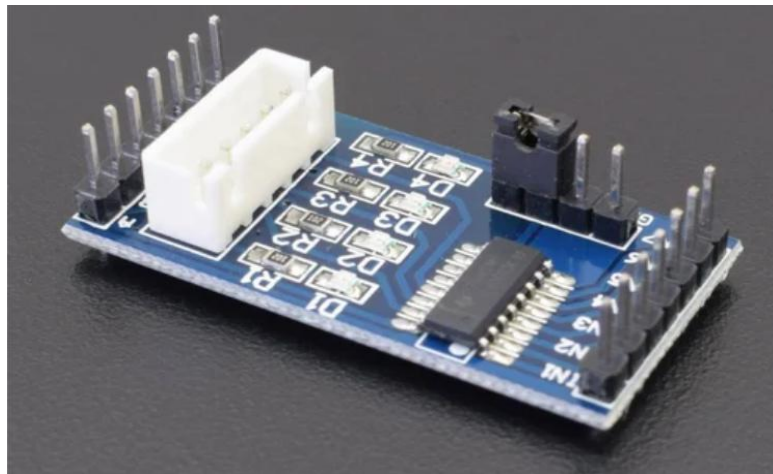
6.5 ULN2803 Driver IC

Another significant hardware in the proposed AI-driven traffic light automation system is the ULN2803 driver IC. It is mainly to power high-current gadgets like motors, relays and lamps that cannot be powered by the microcontroller directly. The proposed system involves the application of the ULN2803 driver IC in order to operate the stepper motor in order to move the camera around to monitor traffic in various directions.

Microcontrollers like the PIC16F877A also use low current output levels that are not adequate to directly power high power devices. When a motor is hard-wired to the microcontroller pins, it might end up destroying the microcontroller because of too much current being taken. In order to circumvent this shortcoming, a driver circuit is applied between the microcontroller and motor. The ULN2803 IC is simply a current amplifier

which enables microcontroller to safely operate high-current devices.

The ULN2803 is an eight-pair Darlington transistor array, a high-voltage and high-current Darlington transistor array as in fig 6.4. The ratio of each pair has the capacity to run loads of up to 500 mA of current and 50 V of voltage. IC also has inbuilt suppression diodes that guard the circuit against voltage spikes produced during switching on or off of the inductive loads like motors. The proposed system will use PIC microcontroller to provide control signals to the ULN2803 driver IC via its output pins. These signals are amplified by the ULN2803 and provide enough current to start the stepper motor. This enables the motor to rotate fluently without interfering with the work of the



microcontroller.

Fig .6.4 ULN2803 Driver IC

The need to use ULN2803 driver IC makes the circuit design easier since it has several driver channels in one package. This also removes the need to have several discrete transistors and decreases the complexity of the circuit as a whole. The other benefit of the ULN2803 IC is that it is reliable and is easy to combine with microcontrollers. IC has been extensively employed in embedded system designs due to the fact that it offers an easy and efficient way of supplying microcontrollers to high-power loads.

Within the suggested traffic monitoring system, the ULN2803 driver IC allows the

stepper motor to turn the camera by which it is possible to observe the traffic situation at different sides of the intersection. The system is capable of collecting the traffic data across a variety of lanes and enhancing the precision of the traffic density estimates by using the option of rotating the camera.

Altogether, the ULN2803 driver IC is an essential component in terms of safe and effective work of the stepper motor. It serves as a point of contact between the low-power control signals of the microcontroller and the increased power needed by the motor and enhances the reliability and performance of the whole traffic monitoring system.

6.6 Stepper Motor

The stepper motor is a significant electromechanical device to be utilized in the proposed AI-driven traffic light automation system shown in fig.6.5. It is mostly applied to turn the Pi Camera to allow the traffic conditions to be observed in various directions of the intersection. The rotation of the camera will allow the system to record the images of several lanes and ensure the complete overview of the traffic conditions.

A stepper motor is a form of electric motor which moves not continuously but in discrete steps, as compared to conventional motors. The motor can move in a high-precision manner and with each move the angular motion is corresponding to a particular step. This linear movement allows accurate control of the motor position hence stepper motors are useful when there is a need to have a great deal of precision with the movement.

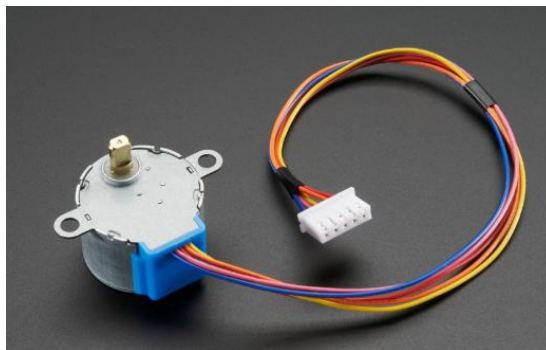


Fig .6.5 Stepper Motor

The PIC microcontroller in the proposed system uses ULN2803 driver IC to control the stepper motor. The microcontroller provides the driver IC with control signals that amplify the signals and provide the current needed to drive the motor. This is followed by the stepper motor rotating based on the series of control signals inputted.

The primary intention of having the stepper motor in this system is to enable the camera to move and keep an eye on traffic in various directions. The intersections normally consist of more than one lane and the view of one direction might be inadequate to draw the full picture regarding the traffic. The system will be able to monitor the incoming traffic in other lanes and collect better data on traffic by rotating the camera.

The next benefit of working with the stepper motor is that it is able to work at small angles. This accuracy provides the ability to adjust the camera in a precise manner to take photos of certain traffic lanes. The motor can be programmed by the system to turn an number of steps to take pictures at predefined points. Embedded systems also have a large usage of stepper motors in that they are very reliable and easy to handle. Microcontrollers may be easily interfaced with them through driver circuits like the ULN2803 IC. The driver circuit will make sure that the motor is supplied with enough current and will not load the microcontroller with a large current.

Besides delivering accurate move, stepper motors have good holding torque which implies that they sustain their motion despite the application of power without constant rotation. This aspect makes the camera to be stable when taking pictures of the traffic. The stepper motor to be employed in the proposed system works on a series of electrical pulses produced by the microcontroller. The motor is made to rotate with a fixed step angle with each pulse. With the ability to control the frequency and the quantity of such pulses, the system is able to control the speed and direction of the motor accurately.

All in all, the stepper motor will be significant in improving the traffic monitoring capacity of the proposed system. The motor allows the rotation of the camera in a way that allows the system to record traffic data in many directions and enhance the performance of traffic density estimation to a greater degree and has more control over the traffic signals.

6.7 Traffic Signal LEDs

The given AI-based traffic light automation system employs traffic signal LEDs as the way to reflect the functioning of the actual traffic lights in the intersection. These LEDs replicate the red, yellow, and green lights used to regulate the flow of vehicles in roads. The Fig 6.6 in the proposed system are coupled to the PIC microcontroller and are turned on according to the decisions made on traffic density decisions by the Raspberry Pi.

The purpose of the traffic lights is to control the traffic in the intersection of the road blocks and ensure other vehicles do not cause accidents. Red signal means that vehicles should stop, the yellow signal means they should be ready to stop, and the green signal is that the vehicles can pass through the intersection. With proper switching of such signals, the traffic movement can be effectively and safely managed.



Fig .6.6 Traffic Signal LEDs

The suggested system has LEDs depicting the functionality of the intelligent traffic control mechanism in place of real traffic lights. They are attached to the PIC microcontroller pin outs by the LED. Every LED corresponds to a given direction of traffic being in a given state (traffic light). These LEDs are switched on and off by means of the microcontroller based on the signal sent to the Raspberry Pi.

When Raspberry Pi perceives vehicles and analyses the traffic density based on the YOLO algorithm, it decides in what lane there is to be more priority. Raspberry Pi

consequently transmits control addresses to PIC microcontroller by means of serial communication. According to these orders, the microcontroller produces the right LED indications. To take an example, in case the system notices that the traffic in a given lane is the heaviest, the microcontroller switches the green LED of that particular lane when the red LEDs of the remaining lanes are on. Upon the expiry of the green signal, the system changes to the yellow signal to signal a transition but before turning the red signal.

The applications of LEDs in electronics systems are immense due to the fact that they use minimal energy and the duration of their life is relatively high. They can also generate bright light consuming low energy and this is why they can be used in demonstrations and signaling purposes. The other benefit of using LEDs is that it has fast switching ability. LEDs can be switched on and off making the microcontroller switch the traffic lights immediately as per the requirements of the systems. This is significant in smart traffic where the timing of signals may require dynamic adjustment of traffic circumstances.

Besides showing traffic lights, LEDs can also be used to give a concrete clue of how the system functions even during testing and demonstration. Visitors can clearly know the automation system about the traffic lights by watching the modifications of the LED lights. In general, traffic signal LEDs will be the result instruments of the suggested system. They show how the intelligent traffic management system manages traffic flow at crossroads by dynamically adjusting the timing of signals depending on the real time traffic flow.

6.8 Crystal Oscillator

The crystal oscillator is a necessary device that is utilized in microcontroller-based systems to make the clock signal stable and accurate. The crystal oscillator would be linked to the PIC microcontroller in the proposed system of AI-headed traffic light automation to produce the clock pulses necessary to be integrated into the system.

A microcontroller needs to have a clock signal to run the instructions and carry out other operations. The clock signal plays a part in defining how fast the microcontroller

processes the information and how the programmed instructions run on the microcontroller. The microcontroller could not take a clock signal to work properly. High precision and stability of these clock pulses are given by the crystal oscillator.

The crystalline oscillator in the proposed system would be linked between the oscillator pins of PIC microcontroller. The oscillator in fig.6.7 produces a constant frequency signal that is the time reference of the microcontroller. This clock signal ensures that the microcontroller does all that it was doing in a known and predictable way. The crystal oscillator accuracy is extremely relevant in order to keep timing in embedded systems on track. Because the PIC microcontroller will be tasked with switching the LEDs of the traffic lights and also the stepper motor, proper timing will give correct timing of traffic lights to be switched. This assists in keeping the right traffic signal sequences of red, yellow and green sequences.



Fig .6.7 Crystal Oscillator

Embedded systems often observe the use of crystal oscillators owing to the fact that these oscillators generate very stable frequency values as opposed to other oscillators. They are created with the use of quartz crystals, which oscillate under the influence of electrical voltage that is introduced into it. These vibrations are generated as a regular clock signal, which the electronic circuits can use.

The other benefit of crystal oscillators is that they are easily reliable and have long life. They can sustain steady frequency output over protracted intervals of time, and diverse conditions of the environment. This is their dependability hence can be used in important control systems like the traffic signal controllers. The crystal oscillator is

further used in synchronizing communication among hardware components besides giving the microcontroller timing signals. In order to ensure that the microcontroller properly interprets commands sent to it by the Raspberry Pi, it is essential to make proper timing so that it does not make mistakes when performing the necessary tasks.

All in all, the crystal oscillator is an instrument that is significant in making sure that the microcontroller works efficiently and concurrently. It helps the traffic signal control system to perform correctly and keep the interval of different signals on time as it promises accurate clock signals.

6.9 Regulated Power Supply

The controlled power supply is also a necessary element of any electronic system as it renders stable and constant voltage needed to properly run the electronic devices. The controlled power supply in the proposed AI-based traffic light automation system is applied to supply electrical power required in all hardware parts of the system such as the Raspberry Pi, PIC microcontroller, LEDs, stepper motor, and other auxiliary circuits.

Electronic components have a certain level of voltage which they need in order to work right. The variation of power supply may lead to mal-function, data errors or even irreversible destruction of electronic parts. This is done to have a controlled power supply to make sure that the system has been fed with a constant voltage level despite any change in the input power source. The controlled power flow normally involves multiple steps that may include transformer, rectifier, filter and voltage regulator. The transformer functionalities include changing the high voltage AC current of the main power source to a lower voltage AC current to be used in electronic circuits. The AC voltage is then transformed in the circuit of the rectifier to direct current (DC), which is needed by the majority of mechanical components.

Even by rectifying, oscillations in the output voltage remain as the so-called ripple voltage. Filter circuits with capacitors are used in order to minimize these fluctuations. These filters line up the voltage waveform and make the DC output to be a more stable value. The last component in the power supply circuit is the voltage regulator. The voltage

regulator maintains the value of the output voltage despite variation in the input voltage or the load conditions. Voltage regulators like the 7805 regulator are also usually also used in many embedded systems in order to supply a constant 5V output needed by the microcontrollers and other digital circuits.

In the proposed system, the regulated power supply makes sure that Raspberry Pi and PIC microcontroller get a steady power supply that can allow them to operate reliably. The circulatory lighting of the traffic lights and driver circuit also depends on the limited power supply to work well.

Maximum significance of the regulated power supply is to guard against the spikes of voltage and electric noise in the system. Constant voltage is used to make sure the microcontroller operates properly and the communication between other hardware components is functioning properly. This has been particularly the case with regard to systems necessitating several components to operate collectively as one. Unstable power to one of the components can impact on the performance of the whole system. The system maintains the performance consistency and long durability by regulating power supply.

In general, the controlled power system is the foundation of the hardware system as it provides stable and dependable electrical power supply to all the components. It makes sure that the smart automation of the traffic lights system is efficient and functional.

CHAPTER 7

SOFTWARE IMPLEMENTATION

7.1 Introduction

The implementation of the software is very important in the intended AI-based traffic light automation. The hardware elements capture the traffic data, and regulate the traffic lights, but the software element does the smart processing needed to support traffic conditions analysis and decision-making. The computer vision, machine learning and embedded system programming are incorporated in the software implementation to produce an intelligent traffic monitoring and signal control system.

The primary computing platform, on which the software is run in the proposed system, is the Raspberry Pi Zero 2W. The Raspberry Pi is powered by a Python process which takes traffic images with the Pi camera and processes them with Image processing and deep learning algorithms. The software constantly analyses the traffic images to identify vehicles and determine the density of traffic at the intersection.

The basic functionality of the software is the use of the YOLO (You Only Look Once) object detection algorithm. YOLO is a real-time object detector that is a deep learning model and can be applied to detect several objects simultaneously. In the present project, the YOLO model is trained on various car models including cars, motorcycles, buses, and trucks. The system can detect the traffic density of each lane by identifying these vehicles in the images of the captured lanes.

Besides detecting vehicles, the software has also applied the logic that is used to control traffic lights. Depending on the traffic intensity observed, the system automatically sets the time of each direction traffic lights. Those lanes that carry more traffic are given longer green lights whereas those that have fewer vehicles are given shorter duration of the green lights.

The software also includes modules of communication that enable the Raspberry Pi to relay control commands to the PIC microcontroller. The transmission of these commands is performed by a serial communication and this allows the microcontroller to turn the traffic lights on accordingly. In general, the software implementation is the wisdom of the traffic monitoring system. It allows the analyzing of the traffic conditions, intelligent decisions and automatic control of the traffic lights without the human influence.

7.2 Programming language-Python

The main programming language in the suggested AI-based traffic light automation system is python. Due to its simplicity, flexibility and large support of libraries, Python finds its application in artificial intelligence and machine learning projects.

A good ecosystem of libraries and frameworks that can be used in computer vision and deep learning is one of the primary reasons why Python was selected. Images Libraries like OpenCV, NumPy, TensorFlow and PyTorch give developers the opportunity to deploy high-level image processing and machine learning algorithms.

The traffic monitoring software, which will be used in the proposed system, is implemented in Python, and it will run on the Raspberry Pi. The Python code reads the images at the Pi Camera and applies the computer vision methods in detecting the vehicles and monitoring traffic density. Python also has simple hardware integration with hardware, including libraries that can be used to communicate with a GPIO and a serial interface. This capability makes the Raspberry Pi able to communicate with the PIC microcontroller and operate traffic lights.

Some other strengths of Python include its readability and easy development. Python is somewhat simple to write, read, and maintain and is suggested to be used in fast development of embedded AI software. All in all, Python is a highly scalable and adaptable platform in terms of implementing smart traffic monitoring systems and combining machine learning algorithms with embedded devices.

7.3 YOLO Object Detection Algorithm

You Only Look Once (YOLO) is a popular deep learning model that is utilized to detect objects in a real-time way. It has wide application in computer vision due to its capability to locate several objects in one image that is very accurate and fast. The YOLO algorithm is based on the division of the input image into a grid of cells. Every grid cell will detect the objects inside of it. The algorithm estimates the bounding boxes of objects and gives confidence scores that show the likelihood of the presence of an object.

Among the most important benefits of YOLO is the fact that it does object detection in only one run through the neural network. This renders the algorithm much faster than conventional object detection algorithms that need to be resorted to repeatedly scanning the image. The YOLO algorithm is applied in the proposed traffic monitoring system to identify vehicles in the traffic scene. The algorithm recognizes vehicles, including cars, buses, motorcycles, and trucks and creates bounding boxes around them.

Through the result of the detection, the system can be able to find the number of vehicles in the intersection in each lane. Traffic density is estimated using this information and decisions on traffic signals control are made with this information. The model of YOLO implemented in this system is being trained on the datasets of pictures of various types of vehicles. The model can identify vehicles in real-time video streams of traffic on the Pi Camera after training to identify the vehicles accurately.

7.4 OpenCV Image Processing

OpenCV (Open Source Computer vision Library) is a popular computer vision library which gives the image processing and video analysis tools. The proposed system applies OpenCV to manipulate the images that the Pi Camera captures and then analyses based on the YOLO detection model.

OpenCV has a number of image data processing functions which include image resizing, color conversion, filtering and frame extracting. These functions assist in ensuring that captured images are ready to be detected as objects through enhancing the

quality of the image and eliminating noise. Frame extraction of the video stream captured by the camera is one of the significant tasks that is done with the help of OpenCV. The Raspberry Pi records video frames and translates the frames into images which the YOLO can process.

OpenCV also enables drawing of bounding boxes on vehicles that are identified and visual representation of the detection performance. This assists in testing the correctness of the vehicle detect system. Combined with the OpenCV and the YOLO model, the system will be capable of detecting and monitoring vehicles and traffic in real time.

7.5 Vehicle Detection Process

The start of the vehicle detection process is when the real-time images of traffic at the intersection are captured by the Pi Camera. Such pictures are relayed to the Raspberry Pi to be processed. Image processing techniques are used to first preprocess the images captured to enhance the image quality. Preprocessing can involve image resizing, brightness and noise removal.

The images are processed afterwards and sent to the YOLO object detection model. The image is analyzed with the help of the model and the vehicles within the traffic scene are detected. The algorithm creates bounding boxes on identified vehicles and classifies them based on the category. Every object that is detected is given a confidence score that represents the likelihood that the detected object is a member of a particular class. The results of the detection are then counted to identify the number of vehicles in the frame. The system uses this information to approximate the traffic density and use it to control the timing of signals.

7.6 Calculation of the density of traffic

Another important aspect of the intelligent traffic management procedure is traffic density calculation. Once vehicles are detected in the captured pictures, the system will tally the number of vehicles within each of the lanes of the intersection.

The number of traffic vehicles is applied to determine the level of traffic density of

a specific lane. It is possible to rate traffic density as low, medium, and high depending on established threshold values. As an illustration, when the quantity of vehicles identified in a lane surpasses some specific number, the system treats that lane to be densely populated. In this scenario, the system will provide a greater green signal time so as to enable more vehicles to cross the intersection.

Conversely, when the amount of vehicles in a lane is very few, the system allocates a shorter signal time. This dynamic change of the signal timings contributes to the better traffic flow and the decreased congestion. The calculation of traffic density is done on an ongoing basis such that the system is able to adjust to the variation in traffic conditions in the real time.

7.7 Serial communication with Microcontroller

This module will involve connecting the system to a microcontroller providing serial communication with the system. The proposed system finds application of serial communication to facilitate communication between the Raspberry Pi and the PIC microcontroller. Through this communication, the Raspberry Pi is able to transmit control commands to the microcontroller depending on the results of the traffic analysis. Serial communication is an easy and efficient way of transmitting data where data is sent bit by bit over a channel of communication. This system has the Raspberry Pi connected to the serial interface to transmit the commands of the traffic lights control to the PIC microcontroller.

Once the Raspberry Pi has done the processing of the traffic images and the calculation of traffic density, it sends out a command as to which traffic signal is to be turned on. This message is sent to PIC microcontroller via the serial communication interface. The microcontroller then takes the command and uses it to operate the traffic light lamps. Depending on the command received, the microcontroller changes the red, yellow and green lights respectively. This communication system enables the system to integrate the computing abilities of the Raspberry Pi with the dependable control of hardware functions of the PIC microcontroller.

CHAPTER 8

CIRCUIT DIAGRAM AND HARDWARE IMPLEMENTATION

8.1 Introduction

The practical base of the proposed AI-driven system of traffic light automation is the circuit diagram and hardware implementation. Whereas in the previous chapters the theoretical concepts, system architecture, and hardware components were mentioned separately, the same chapter dwells upon the relationships between these components with the goal to create a entire working system.

The circuit diagram gives a graphical representation of the electrical links of the different hardware components employed in the system. It displays the connection between devices Raspberry Pi, PIC microcontroller, traffic signal LEDs, stepper motor, ULN2803 driver IC, crystal oscillator and regulated power supply. The circuit diagram is crucial to the understanding of how to install the system and make certain that each of the components fulfills its role.

Hardware implementation Hardware implementation is the actual construction and assembly of the electronic components based on the circuit design. This is done by connecting the components to a development board or a circuit board, and making the correct electrical connections and making sure each device is supplied with the correct voltage and control signals.

The Raspberry Pi Zero 2W will be the central processing unit of the proposed system that will conduct a vehicle identification and image processing with the help of the YOLO deep learning model. The Pi Camera takes live photos of the traffic at the intersection and forwards them to the Raspberry Pi to analyze them. Depending on the measured density of the traffic, the Raspberry Pi creates control signals sent to the microcontroller PCB of the PIC. The PIC microcontroller then uses it to turn the traffic lights LEDs, which are indicative of red, yellow, and green lights of a traffic light. The stepper motor is also controlled by the ULN2803 driver IC connected to the

microcontroller which enables the camera to rotate and view the traffic conditions at other angles.

A correct implementation of the circuit drawing will guarantee a good communication among various hardware units and will allow the system to work properly. The chapter describes the circuit design and gives an elaborate description of the hardware implementation process of the intelligent traffic signal automation system.

8.2 Description of the circuit diagram

The circuit diagram of the proposed traffic light automation system in Fig8.1 shows how the main hardware components used in the system are electrically connected. The

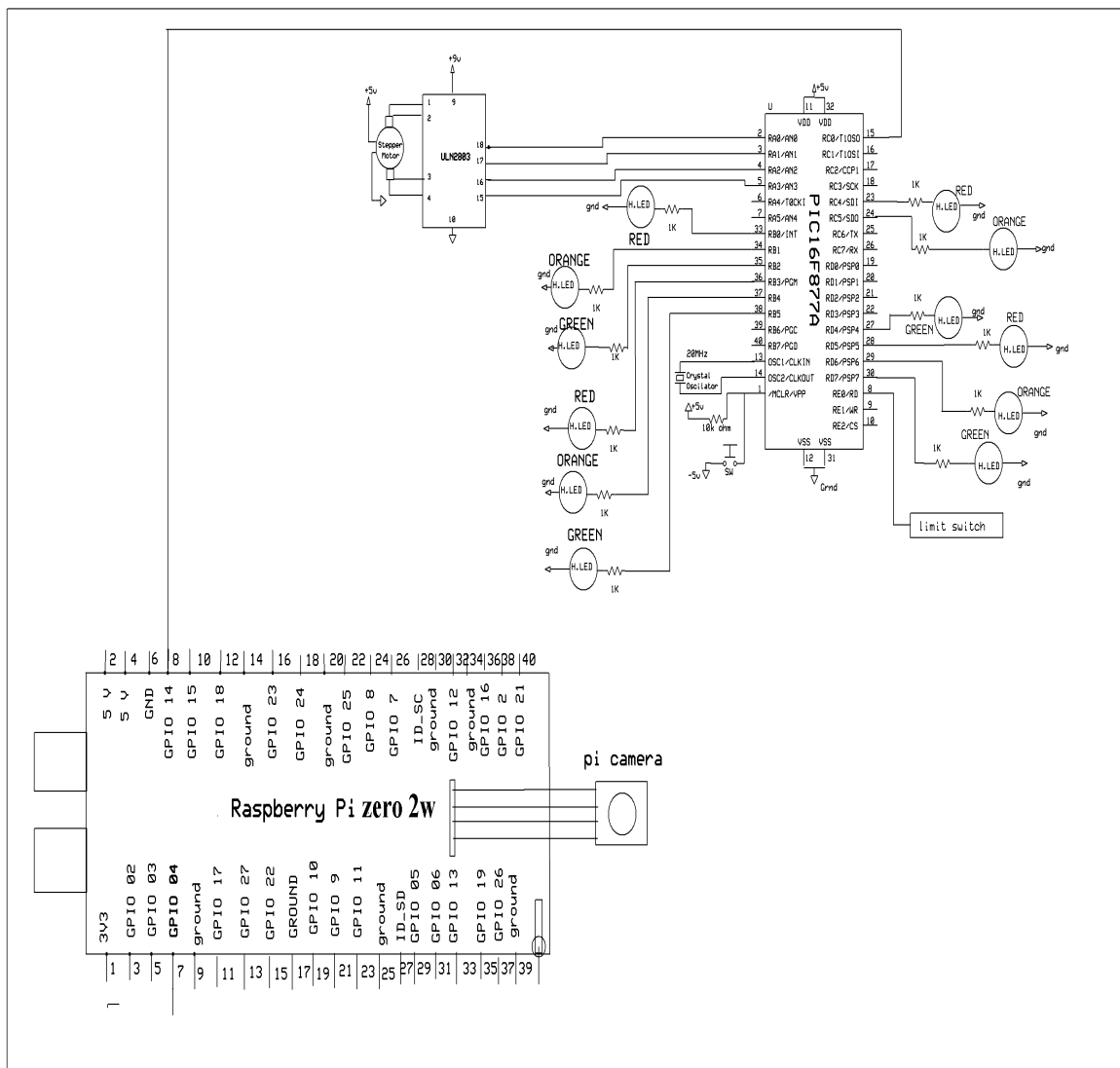


Fig .8.1 Circuit Diagram

circuit is developed in such a way that to have effective communication between the processing unit and control unit and the output devices.

The PIC microcontroller is in the middle of the circuit and it will be used as a controller to the traffic signal. The Raspberry Pi sends commands to the microcontroller which then light the traffic lights depending on the instructions issued by the Raspberry Pi. The microcontroller serves as the linkage between the high level processing unit and the physical traffic signals. The LEDs of the traffic signal can be connected to the output pins of the PIC microcontroller by current limiting resistors. These LEDs signify the red, yellow, and green traffic lights of various directions of traffic movements. Upon receiving a command also sent by the Raspberry Pi, the microcontroller turns on the necessary LED signals to regulate the traffic of vehicles at the intersection.

The camera rotation stepper motor is attached to the microcontroller by ULN2803 driver IC. Due to the fact that the microcontroller is unable to deliver high current needed by the motor, the driver IC is required. The ULN2803 intensifies the output of the control signals of the microcontroller and supplies the current needed to make the motor work.

The crystal oscillator is attached to the oscillator pins of the microcontroller PIC. It creates the clock signal that is needed by the microcontroller to run instructions and carry out operations. Capacitors with the crystal oscillator are used to stabilize the crystal oscillator frequency output and provide proper timing. The circuit design also has a reset circuit. The microcontroller can restart the program when required with the help of the reset switch. This property would be applicable when testing and troubleshooting the system.

The supply part of the circuit supplying power to hardware is also regulated to give all hardware a fixed voltage. The power supply is to make sure that all the parts are supplied with the right operating voltage that allows them to operate stably. In general, the circuit diagram indicates the connection of various electronic components that can make a complete traffic signal automation system. It offers a concise idea of the manner in which signals are relayed among various modules as well as the provision of a system as to how traffic lights are regulated by conditions of traffic.

8.3 Hardware Implementation

Hardware implementation is the process of assembling all the electronic components based on the circuit diagram and is a process that ensures that the system is functioning as desired. To implement the project, it is necessary first to establish the Raspberry Pi and attach it with Pi Camera module.

The Pi Camera is placed in an area that would enable it to have a clear view of the approaching vehicles at the intersection of the traffic. The camera interface port is then attached to the camera. After connecting it, the Raspberry Pi takes the frames of the video captured by the camera and processes them with the use of the traffic monitoring software. The Pic microcontroller will be installed in a microcontroller development board. The development board also has easy access to the input and output pin of the microcontroller and thus can easily connect to LEDs, driver circuit, and other hardware.

The microcontroller output pins are connected to the LEDs of a traffic signal by resistors which restrict the current flowing across the LEDs. These LEDs are the traffic lights towards various directions of the intersection. The microcontroller activates the switching of such LEDs according to the instruction given through the Raspberry Pi.

The ULN2803 driver IC is connected to the stepper motor that is used to rotate the camera. A microcontroller is then linked to the driver IC via the microcontroller output pins. The driver circuit identifies the control signals sent by the microcontroller to the driver IC and so, the motor is turned on and rotates it at an angle.

The crystal oscillator is attached to the microcontroller oscillator pins to give the clock signal that is needed to execute instructions. The oscillator is also linked to capacitors to ensure that the frequency is operated with a stable frequency. All the hardware components are powered by a regulated power supply. Power supply will make sure that the Raspberry Pi, microcontroller and other electronic parts get a constant voltage needed to operate properly. Once a system has been assembled, the system is tested to make sure that all the modules are functioning properly. The Raspberry Pi reads traffic images and transmits commands to the microcontroller which in turn operates the traffic lights. This hardware implementation proves the effective implementation of the

proposed intelligent traffic management system.

8.4 System Integration

The last phase of the hardware implementation process is system integration where all the hardware and software modules are integrated to constitute a complete traffic monitoring system. The integration process makes sure that the various modules interact well and coordinate to bring about intelligent traffic control.

Within the proposed system, the integration will start by interconnection of the processing unit based on the Raspberry Pi and the PIC microcontroller based traffic signal controller. The Raspberry Pi executes the traffic monitoring code which takes pictures of the Pi Camera and identifies the vehicles with the help of the YOLO object detection algorithm. When vehicles are spotted, the Raspberry Pi works out the traffic density of every lane on the intersection. According to the values obtained at the calculations of the density, the system decides what lane to be given precedence and the time that the traffic light ought to stay green.

The Raspberry Pi will then transmit the control command to the microcontroller of PIC through serial communication. The command transmits to the microcontroller which then switches on the relevant traffic light LEDs to control traffic flow. Simultaneously, the microcontroller also transmits control signal to ULN2803 driver IC to turn the stepper motor. The stepper motor moves the camera in such a manner that the situation on the roads in the various directions may be observed.

The microcontroller combined with the other hardware components (Raspberry Pi) allows this coordinated interaction such that the system will always perform this task automatically without the need to manually intervene. The combination of software and hardware parts is necessary to guarantee an effective traffic monitoring and on-the-fly signal control. General, the system integration process shows the interaction between various modules of the suggested traffic management system and their co-operation to avail a smart and adaption traffic control solution.

CHAPTER 9

WORKING PRINCIPLE

9.1 Introduction

The operating model of the suggested AI-based automation system of traffic lights is grounded in the combination of computer vision technologies, artificial intelligence software, and a firewalled hardware control. The system is expected to detect the traffic conditions on the fly and dynamically change the timing of the traffic lights according to the number of vehicles on the intersection.

Traffic jam has been one of the prevailing issues in contemporary cities because the number of motor vehicles has grown fast. The conventional traffic signal systems have fixed times that they use without taking into consideration the real road situation. Consequently, car traffic tends to have unnecessary congestions even in directions where there is a low level of traffic. The intelligent traffic management system proposed shall address this issue by constantly monitoring the traffic flow and adjusting the signal timing to suit the flow.

The system applies a camera-based surveillance system to keep a check on the vehicles coming to the intersection. The camera takes real time pictures of the traffic and transmits them to the Raspberry Pi processing unit. Raspberry Pi processes these pictures with the help of computer vision algorithms and identifies vehicles and computes the traffic density. After the determination of the traffic density, the Raspberry Pi produces control signals which are sent to the PIC microcontroller. The microcontroller will then manipulate the lights of the traffic lights based on the instructions it receives. The system ensures efficient traffic flow by dynamically managing signal timings according to the density of the traffic; it minimizes waiting time in intersections.

The other notable aspect of the system is that it can be used to track several traffic directions through a rotating camera system. The stepper motor will turn the camera in such a way that one can see the traffic conditions in the various lanes one after the other.

This makes the system to capture traffic information in many directions without having to install numerous cameras. In general, the operating principle of the suggested system will consist of traffic surveillance, smart detection of vehicles, traffic density calculation, and the dynamic control of traffic lights. This is a combination of technologies that can make the system an intelligent traffic management solution.

9.2 Traffic Image Capture

The first and one of the most important steps in the functioning of the proposed AI-based system of automation of traffic lights is traffic image capture. Real-time and precise acquisition of images will result in the system having accurate visual data needed to detect vehicles and traffic density. The traffic conditions could not be analyzed effectively without an effective image capture by the system.

A Pi Camera module is utilized as a main image acquisition device in the suggested system. The camera has the Raspberry Pi directly linked to the camera via Camera Serial Interface (CSI) port, which allows the camera to send video data at a high rate to the processing unit. This is a direct interface, which ensures that images captured are delivered effectively with minimum delay.

The camera is placed at a high height at the intersection of the traffic to allow the camera to have a panoramic view of the road and the vehicles that are coming towards the camera. The right location of the camera is very crucial in ensuring that the traffic is monitored properly. The camera might also be too low or at a wrong angle and thus some vehicles might not be captured in the pictures. Thus, the camera will be held in a specific manner to make all the lanes of the road visible. The Pi Camera has the video frames constantly captured at a constant frame rate. Every frame is a picture of the road situation at the particular point in time. These frames are sent to the Raspberry Pi and they are operated on by the traffic monitoring software. The frame rate is chosen in such a manner that it does not overload the processing unit due to an ability to analyze traffic conditions in real time.

The captured frames are pre-processed in a number of steps before they can be used to detect vehicles. These preprocessing tasks are useful in enhancing the quality of the

images that are captured and make them palatable to the object detection algorithms. The image preprocessing can involve enlarging the image, brightness and contrast, changing color formats and image denoising.

The other significant detail in the image capture process is ensuring that there is proper lighting. Traffic surveillance systems should also be able to work in varying light conditions like day time, night time, or when it is cloudy. The camera must have the capability of taking clear shots even when the light conditions vary. In most practical implementations, other features are also added like brightness control and image filter to enhance low-light visibility. The dynamic nature of traffic is ensured as the system monitors the conditions of the traffic by capturing images constantly. The moving vehicles are captured by the camera and the updated frames are availed to the Raspberry Pi. This allows the system to monitor the change in the traffic density and act upon it.

The images of the traffic captured have useful information like vehicle numbers, vehicle location and movement patterns. With the help of these images, the system is able to calculate traffic density and decide which lane has to be given priority at the traffic light. Flexibility is the other benefit of camera based traffic monitoring. Cameras can be easily mounted on the traffic poles or nearby structures unlike the traditional sensor-based systems where one will have to install sensors on the road surface. This saves on the cost of installation and also simplifies maintenance.

Moreover, the camera has the ability to monitor a greater area of space than the traditional sensors. A single camera would be able to capture numerous lanes of traffic at the same time delivering more detailed data on traffic. In sum, the intelligent traffic monitoring system is based on the traffic image capture stage. The nature and correctness of the obtained images directly affect the precision of the vehicle detection and traffic density estimation. The system maintains the ability of traffic lights to be dynamically operated in real time by capturing and transmitting traffic images to the processing unit to be able to manage the flow of traffic.

9.3 Automobile Detection with YOLO

Once the images of the traffic have been registered with the camera, the identification

of the vehicles available in the captured images is the next step in the working process of the proposed system. One of the most significant steps in the intelligent traffic monitoring system is vehicle detection; this is because the system can determine and classify vehicles in real time. Precise vehicle monitoring means that the system will be able to estimate vehicle traffic and control traffic lights respectively.

In the proposed system, vehicle detection will be conducted with the help of the YOLO (You Only Look Once) object detection algorithm. YOLO is a popular object detection model based on deep learning, which has been applied to the fields of computer vision because it is fast and accurate. In contrast with the classical techniques of detecting objects that uses a separate analysis of various segments of an image, YOLO views the whole image by a single pass through a neural network.

YOLO algorithm is based on dividing the input image into a grid of smaller cells. The grid cells have a mandate of identifying objects within their area of interest. The algorithm estimates several bounding boxes of objects in the picture and gives out the confidence values that denote the likelihood of the object. These bounding boxes indicate the position of objects identified like vehicles.

Among the key strengths of YOLO, there is the possibility to detect objects in real-time. The algorithm can simultaneously process the whole image and hence can identify multiple objects in the image without necessarily scanning the image multiple times. This gives YOLO a lot faster compared to most traditional object detection methods.

Applying to the circumstances of the suggested traffic monitoring system, the YOLO model will be trained to identify the various kinds of vehicles that can be commonly encountered on the road. These are cars, motorcycles, buses, trucks among others. In the detection, the algorithm examines the acquired pictures of traffic and determines these vehicles by generating bounding boxes that surround them.

A label is also given to each detected vehicle that denotes the kind of vehicle detected. An example is that the system can recognize the objects detected as car, bus, truck or motorcycle. These tags enable the system to categorize the various kinds of vehicles and learn better how to interpret the traffic make up.

Besides bounding box and labels, the YOLO algorithm also gives the confidence scores of each detected object. The confidence score is the likelihood that the object being identified is of a given category. The detections that are above confidence score set to a predefined threshold are all that is considered to be valid to guarantee reliable vehicle detection.

After the detection of vehicles in the picture, the system derives valuable data out of the detection responses. This contains the quantity of vehicles detected, their location in the picture and their identification. Through this information, the system will be able to calculate the number of vehicles that are on each lane of the road.

The second notable feature of vehicle detection is the possibility to identify more than one vehicle in the traffic jams. The intersection of the traffic usually has numerous vehicles traveling at a time and the detection algorithm should have the capability of resolving every vehicle in the intersections. The high-end features in YOLO enable it to identify several vehicles even in a complicated traffic situation.

The visualization is also done using the results of the detection. The system can outline the detected vehicles of the captured frames by drawing bounding boxes to indicate a visual representation of the detection results. This will assist in ensuring that the algorithm is identifying car traffic in the right way and a very clear picture of the analysis conducted by the system on the traffic scenes.

The quality of the vehicle detection process is determined by a number of factors including image quality, lighting conditions and the training data applied on the YOLO model. With a big database of vehicles images, a trained model can help greatly in enhancing the accuracy of detection, and the performance of the system. In general, vehicle detection phase is very important in the intelligent traffic surveillance system. The system can determine the level of traffic in the traffic in order to reach intelligent conclusions about the control of traffic lights by detecting the vehicles in the traffic accurately. The YOLO deep learning algorithm is fast and reliable in terms of vehicle recognition and thus it can be applied in real-time traffic management.

9.4 Traffic Density Estimation

One of the essential steps of functioning of the proposed AI-driven traffic light automation system is its traffic density estimation. The detection results can be analyzed after the vehicles have been detected in the captured traffic images through the YOLO object detection algorithm and as such, the identification of how congested each lane of each intersection is. This is used to assist the system in making smart choices as far as traffic signal control is concerned.

Traffic density denotes the volume of vehicles that can be found in an area of the road at a given time. It is a significant parameter that is considered to measure traffic conditions and define the way in which the traffic lights should be controlled. Increased traffic density shows that there is high congestion whereas the reduced density shows that the traffic is flowing freely.

Within the framework of the proposed system, the Raspberry Pi interprets the results of the detection made by the YOLO algorithm and the number of vehicles detected during each frame are counted. Detection of each vehicle is defined as a bounding box that indicates the location of the vehicle in the image. Using the position of these bounding boxes, the system identifies the lane of each vehicle.

In order to conduct lane-based traffic density estimation, the image that was captured can be divided into various zones that represent various traffic lanes. Cars that have been identified in each area are counted individually. This enables the system to establish the number of vehicles in every lane of the intersection. The number of vehicles received in each lane is then used to find out the traffic density. The system can establish certain threshold values that will categorize traffic density into a low density, medium density, and high density. As an example, when the amount of vehicles detected on a lane is lower than a specified value, the system assumes that there is a low traffic density. When the number is above the threshold value, the density is taken to be high.

After the density level has been identified, the system takes this information and it determines the manner in which the traffic lights are to be managed. Green signal is granted longer and priority is given to lanes with high traffic rates. This will enable additional vehicles to transit through the intersection, and will ease congestion at that

lane.

Conversely, lanes having a small number of vehicles are awarded shorter periods of the green signal. This is so that the time available to the signal is used efficiently between the various traffic directions. The system enhances the general flow of traffic by timing signals in accordance with the current movement of traffic.

A second reason why traffic density estimation is useful is that it will adjust with a fluctuating traffic scenario. At intersections, the traffic conditions may change quite a lot during the day because of some reasons, including traffic peak hours, accidents, or road closures. The suggested system constantly checks the traffic conditions and re-calculates the values of densities with the new frames being handled.

The dynamism of the density estimation process enables the system to react promptly to sudden changes in the amount of traffic either increasing or decreasing. As an illustration, when a significant number of vehicles reach an intersection through a specific direction, the system has the power to understand that there is an increase in traffic density and automatically readjust the signal timing to facilitate the extra cars.

Along with the counting of the vehicles, the advanced methods of traffic density estimations can also take into account the speed of the vehicle, type of the vehicle, and the distance between the vehicles. These aspects may bring more information in the traffic conditions and enhance the precision of traffic analysis. All in all, the traffic density estimation is an important element of the intelligent traffic management system. The system will be able to make decisions based on traffic signal control by estimating the number of vehicles in each lane correctly. This assists in decongestion, reduction of waiting periods and overall efficiency of traffic flow in intersections.

9.5 Traffic Signal Control

The last step of the working process in the proposed AI-driven traffic light automation system is the traffic signal control. Once the system takes a snapshot of the traffic and identifies vehicles through the YOLO algorithm and calculates the density of the traffic in each of the lanes, the traffic lights should then be controlled accordingly.

This phase is in charge of controlling traffic at the intersection concerning the analyzed traffic conditions.

The Raspberry Pi and the PIC microcontroller coordinate to give control to traffic lights in the proposed system. The Raspberry Pi is the central processing unit which runs image analysis and estimation of the density of traffic. After the density values have been figured out, the Raspberry Pi decides the lane that should have priority and creates control signals to the traffic lights.

Under serial communication, the control obscure that is created by Raspberry Pi is sent to PIC microcontroller. Serial communication will be used to help communicate with the microcontroller efficiently and reliably with Raspberry Pi. These commands are sent to the microcontroller which interprets them to turn on the traffic light LEDs on.

The light switching of the traffic lights is done by the PIC microcontroller. The yellow and red and green LEDs that signify the traffic lights are attached to the microcontrollers output pin. Depending on the instructions delivered by the Raspberry Pi, the microcontroller will turn the right lights on or off to control the motion of vehicles.

In case one lane contains the most traffic, the system prioritizes the lane. For that direction, the microcontroller switches the green signal on, and the vehicles on that lane can go through the intersection. Meanwhile, the red lights will still be on the other lanes to ensure that cars do not move at the same time and cause accidents.

The green signal lasts according to the amount of traffic that the system has detected. Lanes that have more traffic are allocated more time in green lights to enable more vehicles to pass out the crossing point. On the other hand, green signal time is given a shorter period to lanes with less traffic density.

The green signal is followed by a brief period of the yellow signal issued by the system after the time of the green signal. The yellow signal is issued to alert the drivers, that the signal is going to shift. This changeover period assists the drivers in making preparations to halt their car. When the time of the yellow signal is over, the microcontroller turns the red signal on with the same lane and the green signal to another

lane depending on the priority according to which the system was built. This is done in cyclic fashion such that every lane is given a relevant signal duration depending on the density of traffic. The other significance in traffic signal control is the need to ensure the proper synchronization of various traffic signals.

9.6 Camera Rotation Mechanism

In the AI-based traffic light automation model, the traffic situation in different directions should be monitored to analyze the traffic correctly and promptly control traffic signals. One camera standing in a fixed location might fail to provide traffic related information in all directions of a busy intersection. To defeat this shortcoming, the system is designed in such a way that it has a camera rotation system with the help of a stepper motor.

The stepper motor will enable the camera to turn at fixed angles in order to take a shot of the cars coming to the intersection in various directions. The system can be used to check the traffic conditions in various lanes by rotating the camera at regular intervals without necessarily having several cameras to check the traffic conditions. This will minimize the total system cost and still offer an extensive traffic surveillance.

The stepper motor is linked to PIC microcontroller by the ULN2803 driver IC. The motor and the microcontroller connect to each other via the driver IC. As the microcontroller is not capable of delivering the high current needed to drive the motor, the ULN2803 driver IC amplifies the control signals produced by the microcontroller and delivers enough current to run the motor.

The microcontroller transmits a series of electrical pulses to the driver IC, which in turn causes the coils on the stepper motor to be switched on in a given sequence. When this series of pulses is fed to the motor it causes the motor to turn step by step. Every pulse is related to a tiny angular change which is referred to as a step. The microcontroller can also regulate the direction of the motor rotation and its speed accurately by regulating the amount and order of pulses.

The stepper motor has the camera attached to the shaft in the proposed system. As

the motor rotates, the camera also rotates as the motor is rotating. To capture the camera in various angles of the intersection, the microcontroller can be programmed to move the camera by a specific angle like 90 degrees to obtain the image.

As an example, the camera could first of all capture the pictures of vehicles moving in the north direction. Once the captured frames are processed and the level of traffic density of a certain direction is estimated by the microcontroller, the camera is turned towards the east. This is reiterated to other directional directions of the intersection. The periodical nature of the rotation process is done in order to ensure that the system receives fresh traffic data in all directions. This will make the traffic monitoring system have a full view of the intersection and therefore give an accurate estimate of the traffic density of each lane.

The next benefit of a stepper motor as the solution of camera rotation is the possibility to move it precisely. The stepper motors have a distinct rotation unlike the traditional motors and this feature enables the system to move the camera to its desired position. This accuracy makes the camera to have photos taken at the right angle each time the camera turns. The traffic monitoring system is made flexible as well through the camera rotation mechanism. In case of the variation in the traffic pattern or when the system requires to observe more.

CHAPTER 10

RESULTS AND ANALYSIS

10.1 Introduction

The analysis chapter gives the performance analysis of the offered AI-based traffic lights automation system. The chapter describes the functionality of the developed system under the conditions of a real-time traffic monitoring and proves the efficiency of intelligent traffic management approach.

The primary goal of this chapter is to examine the output of the system and determine as well as confirm that the system is functioning according to the intended functions. The outcomes of the system are vehicle detection findings, estimation of the traffic density, and automatic traffic signal control in accordance with the identified traffic conditions.

The proposed system would have the Raspberry Pi tracking the real-time traffic images with the Pi Camera and identifying the vehicles based on the YOLO object detection algorithm. Results of the detection are thereafter used to determine the estimate of various lanes traffic density. Using this information the system reacts dynamically to adjust the timings of traffic lights to enhance traffic smooth flow.

The findings of the system show how the proposed traffic monitoring system can distinguish vehicles and manage traffic lights properly. Some of the parameters using which the performance of the system can be measured are the accuracy of the system in detecting, the speed of response and the ability of the system to decrease traffic congestion.

10.2 Vehicle Detection Results

One of the most significant tasks of the proposed intelligent traffic monitoring system is vehicle detection. Carrying out object detection on vehicles in the traffic pictures is done using the YOLO object detection algorithm. The Pi Camera is used to record the scene of traffic by recording video frames of the system while it is operating.

YOLO detection model processed the frames on the Raspberry Pi. The algorithm processes the frames and detects vehicles by creating bounding boxes of objects detected.

The results of the detection clearly shown in the fig.10.1 and fig.10.2. Every identified vehicle is tagged based on its category, i.e. car, bus, truck, or motorcycle. Bounding boxes are used to illustrate the results of the detection and ensure that the algorithm succeeds in detecting vehicles of the traffic scene correctly.



Fig .10.1 Detection of Emergency vehicle



Fig .10.2 Detection of cars

The system used in the detection can find several vehicles in one frame. The YOLO algorithm is able to detect and classify multiple vehicles even in the scenario when there are many of them within very close distance. It has factors that determine whether vehicles are detected correctly and these factors include image quality, lighting conditions, and camera positioning. The system had a reliable detection performance in most of the test cases and was capable of detecting vehicles. This is because these results of detection can be used to come up with the input needed to measure the density of traffic, and to ensure dynamically controlled traffic lights.

10.3 Traffic Density Analysis

The system analyses traffic density after identification of vehicles in the captured frames. The steps involved are to count out the number of vehicles in each lane and also give an estimation as to the degree of congestion at the intersection. The Raspberry Pi is the one which analyzes the bounding boxes created by the YOLO algorithm and calculates the count of the vehicles in every part of the image. Every region is attached to a certain traffic lane proceeding to the intersection.

Depending on the number of vehicles, the system gives the traffic density various grades including low density, medium density and high density traffic density. The lanes which have high number of vehicles are defined as high-density lanes and are given priority during traffic light management.

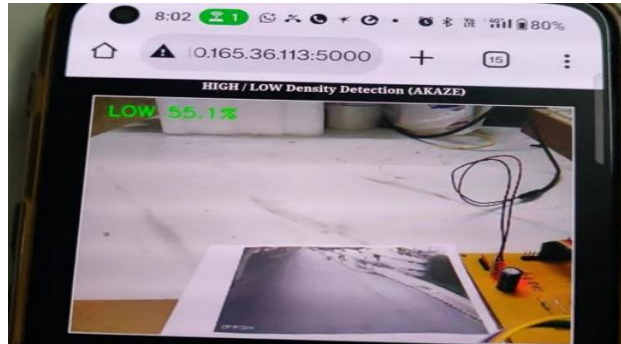


Fig .10.3 Low traffic density with green signal duration of 6s



Fig .10.4 High traffic density with green signal duration of 12s

The traffic density analysis shown in fig.10.3 and fig.10.4 enables the system to know the traffic condition in real time and make intelligent decisions on signal timings. Considering this as an example, in case one lane contains very many vehicles compared to another, the system will provide that one with a longer time of green light. The variable timing of the signals depending on the amount of traffic arriving at the intersection assists in minimizing congestion and positively influences the traffic flow in general.

10.4 Traffic Signal Operation

The PIC microcontroller is used to manage the traffic signals operation on the basis of the commands sent by the Raspberry Pi. The Raspberry Pi, on reaching out to the microcontroller via serial interchange, issues control commands to the microcontroller to

adjust signal timing depending on the density of the traffic. The PIC microcontroller then sets the respective LEDs of traffic lights. The LEDs signify the red color, yellow color, and green color traffic lights utilized to control the traffic flow. Once given the green signal, the vehicles within that lane are given a chance to pass through the intersection whereas other lanes are kept in the red signal position. Once the duration of the green signal is over the yellow signal is then activated, but only briefly, to signal a transition after which the signal turns into red. The system is shown in Fig.10.5 and fig.10.6.

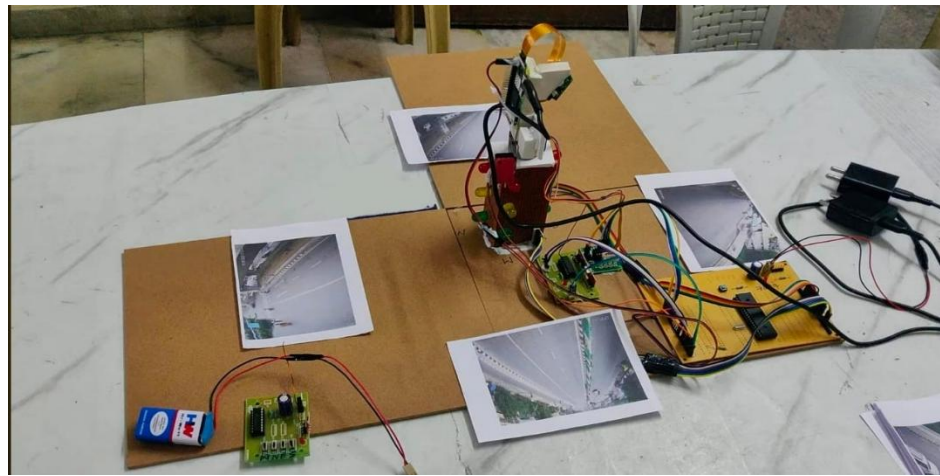


Fig .10.5 Hardware setup of traffic system

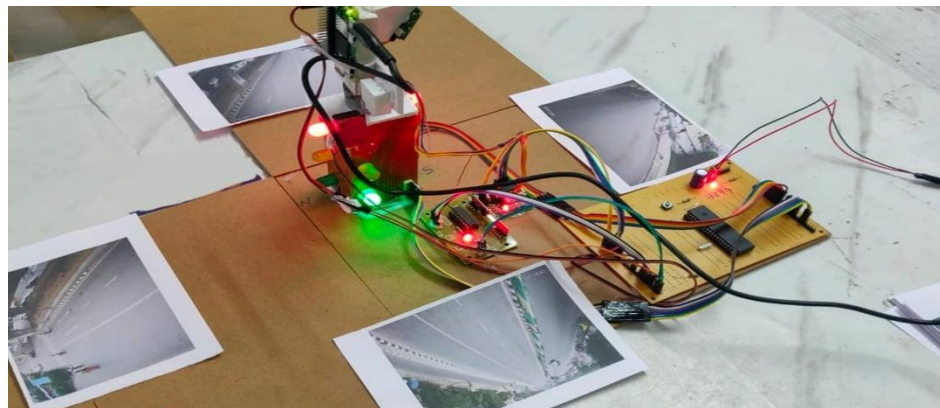


Fig .10.6 Working model of traffic system

The Microcontroller keeps updating the state of the signals according to the commands made by the Raspberry Pi. This is an automated signal control system that makes the traffic lights adaptable to the real time traffic conditions. The way the traffic signal works proved that the system is able to manage traffic lights and control traffic at the intersection.

10.5 System Performance Evaluation

The intelligent traffic monitoring system can be assessed in relation to a number of factors which include, among others, detection accuracy, response time and improvement of the traffic flow shown in fig.10.7,fig.10.8,fig.10.9,fig.10.10. The YOLO object detector algorithm proved to be very accurate in identifying vehicles in the frames that were taken. In most of the test cases, the system was able to identify several vehicles at a given time and categorize them appropriately.

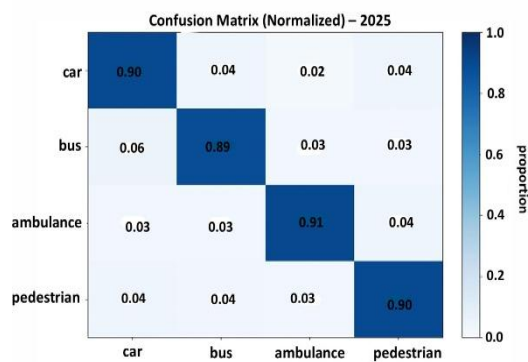


Fig .10.7 Confusion Matrix

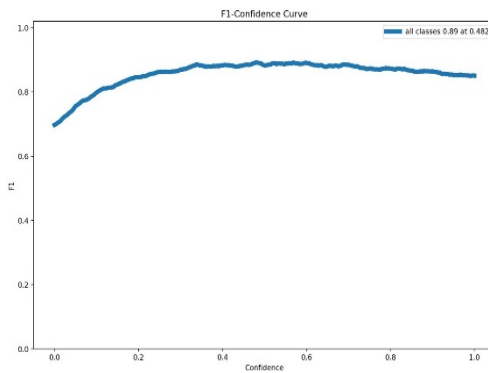


Fig .10.8 F1-Confidence Curve

Another performance factor is the response time of the system. As the YOLO algorithm works with images on the fly, it means that the system will be capable of analyzing traffic conditions and adjusting signal timings on a quick basis.

The other significant performance aspect is the ability of the system to minimize traffic congestion.

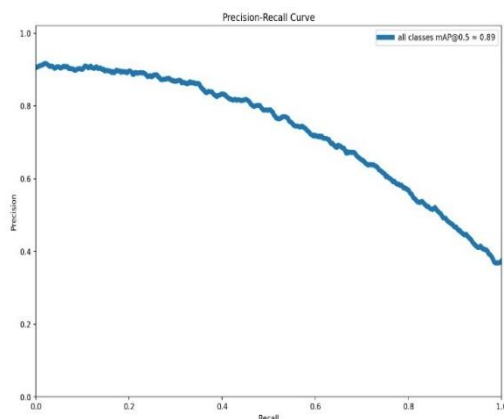


Fig .10.9 Precision-Recall Curve

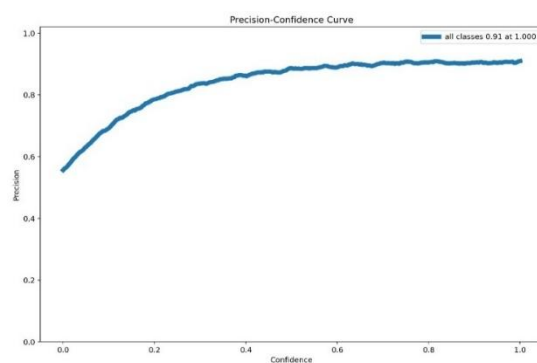


Fig .10.10 Precision-Confidence Curve

The system, which provides the lanes with the most traffic with longer green lights by dynamically changing the signal times according to the number of traffic, is therefore able to cater to the concerns of the majority.

This assists in minimizing the waiting time and traffic. On the whole, the obtained outcomes of the system prove that the offered AI-based traffic light automation system could work as an efficient tool to monitor traffic situation and manage traffic lights intelligently.

CHAPTER 11

CONCLUSION

The proposed AI-based traffic light automation system will overcome these shortcomings by bringing in a smart traffic control mechanism that will dynamically regulate traffic lights in relation to the traffic conditions in real time. It is a system that combines computer vision technology, deep learning algorithms, and hardware of embedded systems to develop a smart traffic monitoring/ signal control system.

The results of the experiment conducted as the implementation system showed that the suggested method can effectively identify the vehicle, traffic density, and manage traffic lights. Fast and precise detection of vehicles is guaranteed by the use of the YOLO deep learning algorithm, which makes the system appropriate to use in real-time traffic monitoring projects.

There are various benefits that come with the implementation of an intelligent traffic signal control system than the traditional fixed-time traffic control system. It minimizes unnecessary waiting time in the cross-roads, optimizes the movement of the traffic, and assists in minimizing the traffic jam. Moreover, the system helps to decrease the fuel usage and decrease the environmental pollution by cutting the time of vehicles standing at traffic lights.

In general, the offered AI-based traffic lights automation system proves the potential of computer vision and artificial intelligence technology in enhancing the control of traffic in cities. The system integrates real time traffic monitoring with dynamic control of the signals which is a good solution to the problem of controlling traffic at a busy intersection. The project emphasizes the significance of the combination of latest technologies, including machine learning, image processing, and embedded systems, to create smart transportation solutions. Such systems with additional improvements and widespread implementation can be of great importance in the construction of smarter and efficient transportation infrastructure in the future smart cities.

REFERENCES

- [1] R. R. Maurya, P. N. Verma, and N. S. Kumar, “Density-Based Traffic Signal Control Using Image Processing,” *YMER Journal*, vol. 22.
- [2] U. E. Prakash and A. Thankappan, “Density Based Traffic Control System Using Image Processing,” in *Proc. Int. Conf. Emerging Trends and Innovations in Engineering and Technological Research (ICETIETR)*, Kollam, India, 2018.
- [3] S. S. P. Moka, D. H. Dande, and P. V. N. Santhosh, “Real-Time Density-Based Traffic Surveillance System Integrated with Acoustic-Based Emergency Vehicle Detection,” in *Proc. 4th Int. Conf. Computer, Communication and Signal Processing (ICCCSP)*, Chennai, India, 2020.
- [4] D. Zabala-Blanco, D. Gonzalez-Caballero, and A. Orozco-Alzate, “Traffic Sign Detection and Recognition Using YOLO Object Detection Algorithm: A Systematic Review,” *Mathematics*, vol. 12, no. 2, p. 297, Jan. 2024.
- [5] K. Sai Venu Prathap, M. Anusha, and S. Jyothi, “Intelligent Traffic Light System Using YOLO,” in *Innovations in Signal Processing and Embedded Systems*.
- [6] A. S. Pawar and R. K. Sharma, “Smart Traffic Control System Using IoT and Computer Vision,” *Int. J. Advanced Research in Computer and Communication Engineering*, vol. 9, no. 4, 2023.
- [7] T. Nguyen and L. Tran, “A Real-Time Vehicle Detection System Using YOLOv8 for Traffic Flow Analysis,” *IEEE Access*, vol. 11, pp. 104512–104523, 202
- [8] S. Mahima, Shree Lakshmi R. Patil, Dayasagar K. V., and Divya B. M., “Real Time Image Processing Based Intelligent Traffic Control System,” *International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE)*, 2021.
- [9] S. Nandhini, M. R. Magesh Kumar, M. Mownika, K. Ranganayagi, and R. J. Pavithra,

“Smart Traffic Surveillance System with Adaptive Traffic Control Signal using YOLO,” International Journal of Engineering Research & Technology (IJERT), vol. 13, issue 05, 2023.

- [10] P. R. Boppuru, P. K. Kukatlapalli, C. R. Chowdary, and J. Hussain, “AI-Based YOLO V4 Intelligent Traffic Light Control System,” Journal of Automation, Mobile Robotics and Intelligent Systems, vol. 16, no. 4, pp. 53–61, 2022.
- [11] Vitthal B. Kamble, Onkar N. Mundhe, Chaitanya M. Walunjkar, and Gaurav A. Kale, “AI-Driven Smart Traffic Management System: An Adaptive Approach Using YOLO and OpenCV,” International Journal of Multidisciplinary on Science and Management (IJMSM), vol. 2, no. 2, pp. 66–72, 2025.
- [12] Trupti Shah, Anup Ingle “Smart Traffic: Integrating Machine Learning, and YOLO for Adaptive Traffic Management System,” International Journal of Intelligent Systems and Applications in Engineering (IJISAE), vol. 12, no. 12s, 2024.
- [13] Nyatha Vinay Kumar, Vijay Gugulothu, Samala Shravan Kumar & Banda Srujan Kumar, “Intelligent Traffic Signal Control using Deep Learning and Reinforcement Learning,” Journal of Engineering Sciences, vol. 16, issue 04, 2025.
- [14] J. Kosasi, Z. A. L. Sihombing, and A. M. Husein, “YOLO-Based Vehicle Detection: Literature Review,” Journal of Computer Networks, Architecture and High Performance Computing, vol. 6, no. 3, pp. 1384–1389, July 2024.
- [15] Cheruku Sindhu, Arram Sharanya, and K. Venkatesh Sharma, “AI-Driven Smart Traffic Control System Using CNN and YOLO for Real-Time Urban Mobility Optimization,” Frontiers in Collaborative Research, vol. 3, no. 1, 2025.
- [16] Raian Rahman, Zadid Bin Azad, and Md. Bakhtiar Hasan, “Densely-Populated Traffic Detection using YOLOv5 and Non-Maximum Suppression Ensembling,” arXiv preprint, 2021.
- [17] C. Padmaja and T. K. Kumar, “Integrating Explainable AI in IoT for Enhancing System Transparency and Performance,” in Proceedings of the International Conference on Intelligent Systems and IoT Applications, 2025.

- [18] B. V. P. Babu and D. Meghana, “IoT-Enabled Plant Management: A Solution for Better Automation,” in Lecture Notes in Electrical Engineering, Singapore: Springer, 2025.
- [19] C. Paseddula and S. V. Gangashetty, “Acoustic Scene Classification Using Various Feature Extraction Techniques,” Circuits, Systems, and Signal Processing, vol. 44, no. 10, pp. 1–12, 2025.
- [20] K. V. Dhana Lakshmi, G. N. Mallika, and E. Sruthi, “Power Saving Mechanism for Street Lights System Using IoT,” in Lecture Notes in Electrical Engineering, Singapore: Springer, 2025.
- [21] C. Padmaja, “IoT Enabled Smart Emergency Response System,” in Lecture Notes in Electrical Engineering, Singapore: Springer, 2025.